IEC 61076-3-104

Edition 3.0 2017-05

INTERNATIONAL STANDARD

colour inside

Connectors for electrical and electronic equipment – Product requirements – Part 3-104: Detail specification for 8-way, shielded free and fixed connectors for data transmissions with frequencies up to 2 000 MHz

IEC 61076-3-104:2017-05(en)

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FOREWORD

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International Standard IEC 61076-3-104 has been prepared by subcommittee 48B: Electrical connectors, of IEC technical committee 48: Electrical connectors and mechanical structures for electrical and electronic equipment.

This third edition of IEC 61076-3-104 cancels and replaces the second edition, published in 2006, and constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

a) the title has been changed to incorporate transmissions with frequencies up to 2 000 MHz;
b) the drawings of some styles have been corrected for clarification;
c) Figures 23 and 24 have been updated;

d) Figure 3 has been updated to include reference dimensions and dimensional format changes;
e) the dimensions of Figure 7 have been updated;
f) the type designation and ordering information has been removed for consistency with the most updated sectional specification;
g) the test schedule was updated to include appropriate IEC 60512 Test Nos;
h) the electrical performance requirements have been revised for 2 GHz level;
i) interchangeability information has been added for performance categories.

The text of this International Standard is based on the following documents:

FDIS	Report on voting
48B/2560/FDIS	48B/2565/RVD

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts of the IEC 61076 series, published under the general title *Connectors for electronic equipment – Product requirements*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "http://webstore.iec.ch" in the data related to the specific document. At this date, the document will be

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- replaced by a revised edition, or
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INTRODUCTION

The International Electrotechnical Commission (IEC) draws attention to the fact that it is claimed that compliance with this document may involve the use of a patent concerning connectors given in 4.2.2 and 4.2.4.

The IEC takes no position concerning the evidence, validity and scope of this patent right.

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CONNECTORS FOR ELECTRICAL AND ELECTRONIC EQUIPMENT – PRODUCT REQUIREMENTS –

Part 3-104: Detail specification for 8-way, shielded free and fixed connectors for data transmissions with frequencies up to 2 000 MHz

1 Scope

This part of IEC 61076 establishes uniform specifications, type testing requirements for 8-way, shielded free and fixed connectors for data transmissions with frequencies up to 2 000 MHz, and used as category 7_A connectors in class F_A cabling systems specified in ISO/IEC 11801-1. It contains all test methods and sequences, severity and preferred values for dimensions and characteristics.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60050-581, *International Electrotechnical Vocabulary – Part 581: Electromechanical components for electronic equipment*

IEC 60068-1, *Environmental testing – Part 1: General and guidance*

IEC 60169-15, *Radio-frequency connectors – Part 15: R.F. coaxial connectors with inner diameter of outer conductor 4.13 mm (0.163 in) with screw coupling – Characters impedance 50 ohms (Type SMA)*

IEC 60352-2, *Solderless connections – Part 2: Crimped connections – General requirements, test methods and practical guidance*

IEC 60352-3, *Solderless connections – Part 3: Solderless accessible insulation displacement connections – General requirements, test methods and practical guidance*

IEC 60352-4, *Solderless connections – Part 4: Solderless non-accessible insulation displacement connections – General requirements, test methods and practical guidance*

IEC 60352-5, *Solderless connections – Part 5: Press-in connections – General requirements, test methods and practical guidance*

IEC 60352-6, *Solderless connections – Part 6: Insulation piercing connections – General requirements, test methods and practical guidance*

IEC 60352-7, *Solderless connections – Part 7: Spring clamp connections – General requirements, test methods and practical guidance*

IEC 60512-1-1, *Connectors for electronic equipment – Tests and measurements – Part 1-1: General examination – Test 1a: Visual examination*

IEC 60512-1-2, *Connectors for electronic equipment – Tests and measurements – Part 1-2: General examination – Test 1b: Examination of dimension and mass*

IEC 60512-2-1, *Connectors for electronic equipment – Tests and measurements – Part 2-1: Electrical continuity and contact resistance tests – Test 2a: Contact resistance – Millivolt level method*

IEC 60512-2-5, *Connectors for electronic equipment – Tests and measurements – Part 2-5: Electrical continuity and contact resistance tests – Test 2e: Contact disturbance*

IEC 60512-3-1, *Connectors for electronic equipment – Tests and measurements – Part 3-1: Insulation tests – Test 3a: Insulation resistance*

IEC 60512-4-1, *Connectors for electronic equipment – Tests and measurements – Part 4-1: Voltage stress tests – Test 4a: Voltage proof*

IEC 60512-6-4, *Connectors for electronic equipment – Tests and measurements – Part 6-4: Dynamic stress tests – Test 6d: Vibration (sinusoidal)*

IEC 60512-9-1, *Connectors for electronic equipment – Tests and measurements – Part 9-1: Endurance tests – Test 9a: Mechanical operation*

IEC 60512-9-2, *Connectors for electronic equipment – Tests and measurements – Part 9-2: Endurance tests – Test 9b: Electrical load and temperature*

IEC 60512-11-4, *Connectors for electronic equipment – Tests and measurements – Part 11-4: Climatic tests – Test 11d: Rapid change of temperature*

IEC 60512-11-7, *Connectors for electronic equipment – Tests and measurements – Part 11-7: Climatic tests – Test 11g: Flowing mixed gas corrosion test*

IEC 60512-13-2, *Connectors for electronic equipment – Tests and measurements – Part 13-2: Mechanical operation tests – Test 13b: Insertion and withdrawal forces*

IEC 60512-15-6, *Connectors for electronic equipment – Tests and measurements – Part 15-6: Connector tests (mechanical) – Test 15f: Effectiveness of connector coupling devices*

IEC 60512-26-100, *Connectors for electronic equipment – Tests and measurements – Part 26–100: measurement setup, test and reference arrangements and measurements for connectors according to IEC 60603-7 – Test 26a to 26g*

IEC 60512-28-100, *Connectors for electronic equipment – Tests and measurements – Part 28-100: Signal integrity tests up to 1 000 MHz on IEC 60603-7 and IEC 61076-3 series connectors – Test 28a to 28g*

IEC 61076-1:2006, *Connectors for electronic equipment – Product requirements – Part 1: Generic specification*

IEC 61076-3:2008, *Connectors for electronic equipment – Product requirements – Part 3: Rectangular connectors – Sectional specification*

IEC 61156 (all parts), *Multicore and symmetrical pair/quad cables for digital communications*

IEC 61156-2, *Multicore and symmetrical pair/quad cables for digital communications – Part 2: Symmetrical pair/quad cables with transmission characteristics up to 100 MHz – Horizontal floor wiring – Sectional specification*

IEC 61156-3, *Multicore and symmetrical pair/quad cables for digital communications – Part 3: Work area cable – Sectional specification*

IEC 61156-4, *Multicore and symmetrical pair/quad cables for digital communications – Part 4: Riser cables – Sectional specification*

IEC 61156-5, *Multicore and symmetrical pair/quad cables for digital communications – Part 5: Symmetrical pair/quad cables with transmission characteristics up to 1 000 MHz – Horizontal floor wiring – Sectional specification*

IEC 61156-6, *Multicore and symmetrical pair/quad cables for digital communications – Part 6: Symmetrical pair/quad cables with transmission characteristics up to 1 000 MHz – Work area wiring – Sectional specification*

IEC 61156-7, *Multicore and symmetrical pair/quad cables for digital communications – Part 7: Symmetrical pair cables with transmission characteristics up to 1 200 MHz – Sectional specification for digital and analog communication cables*

IEC 61156-9, *Multicore and symmetrical pair/quad cables for digital communications – Part 9: Cables for channels with transmission characteristics up to 2 GHz – Sectional specification*

IEC 61156-10, *Multicore and symmetrical pair/quad cables for digital communications – Part 10: Cables for cords with transmission characteristics up to 2 GHz – Sectional specification*

IEC 62153-4-12, *Metallic communication cable test methods – Part 4-12: Electromagnetic compatibility (EMC) – Coupling attenuation or screening attenuation of connecting hardware – Absorbing clamp method*

ISO/IEC 11801-1[1], *Information technology – Generic cabling for customer premises – Part 1: General requirements*

ISO 1302, *Geometrical Product Specifications (GPS) – Indication of surface texture in technical product documentation*

Recommendation ITU-T K.44:2012, *Resistibility test for telecommunication equipment exposed to overvoltages and overcurrents – Basic recommendation*

EN 50289-1-14, *Communication cables – Specifications for test methods – Part 1-14: Electrical test methods – Coupling attenuation or screening attenuation of connecting hardware*

3 Technical information

3.1 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 61076-1, IEC 60512-1 and IEC 60050-581 and the following apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at http://www.electropedia.org/

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1 To be published.

- ISO Online browsing platform: available at http://www.iso.org/obp
- Type: Connectors within a particular subfamily such as a multicontact connector with one, two or four pairs.
- Range: The housing (shell) sizes and contact arrangements within a type. For example, a housing containing one, two or four pairs.
- Style: A particular connector within a type, for example fixed panel, PCB or free connector.
- Variant: Variations within a type, style or range, and Interchangeability.

3.1.1
AACR-F
attenuation to alien (exogenous) crosstalk ratio at the far-end AACR-F
difference in dB, between the alien far-end crosstalk loss from a disturbing pair of a channel and the insertion loss of a disturbed pair in another channel

3.2 Levels of compatibility

3.2.1 Interchangeability level

The connectors shall be intermateable, interoperable and backwards compatible with IEC 61076-3-104 Category 7_A and IEC 61076-3-104 Category 8.2 connectors with the limitation that transmission performance will be to that of the lower performing connector.

The connectors shall be interchangeable insofar as the intermateabilty and interoperability requirements herein, which ensure that this characteristic applies to mated connectors even when individual connector halves are from different sources. The mechanical and electrical characteristics shall be met whatever the source of the connector used. Conformity is achieved by meeting the requirements of 3.2.2.

NOTE The plug/socket interface can be constructed so as to permit the use of multiple modules, e.g. 2 × 2 pairs or 4 × 1 pair plugs mated directly with a single 4 pairs socket.

3.2.2 Intermateabilty

Intermateabilty shall be ensured by applying the “Go” and “No-Go” gauge requirements herein, and by meeting the dimensional requirements herein.

3.2.3 Interoperability

Interoperability of different IEC 61076-3-104 Category 7_A connectors shall be assured by compliance with all transmission requirements when the connector is mated with the respective “test” connector as described in the normative annexes of this standard.

Interoperability of different IEC 61076-3-104 Category 8.2 connectors shall be assured by compliance with all transmission requirements when the connector is mated with the respective “test” connector as described in 5.4 of this standard.

The connector types are interoperable with the limitation that the transmission performance will be identical to that of the lower performing connector.

3.3 Groups of related connectors

A group of related connectors shall be defined as a group of connectors within a family having common features. Typical examples of this will be the same connector type and range, but different style.

NOTE The IEC 61076-3-104 is a group of related connectors, which are covered by this detailed specification.

3.4 Classification into climatic categories

Classification into climatic categories shall be as specified in 5.3.

3.5 Clearance and creepage distances

Clearance and creepage distance shall be as specified in 5.4.1.

3.6 Current carrying capacity

Current carrying capacity shall be as specified in 5.4.3.

3.7 Marking

3.7.1 General

The marking of the connector and package shall be in accordance with 2.6 of IEC 61076-3:2008.

3.7.2 Performance marking

IEC 61076-3-104 Category 7_A connecting hardware shall have no performance marking on the housing.

IEC 61076-3-104 Category 8.2 connecting hardware shall have a performance mark on the housing visible during installation. The marking shall be:

“Cat 8.2” for IEC 61076-3-104 Category 8.2 components.

NOTE Performance markings are an addition to the markings designated in IEC 61076-3. These markings are not to replace safety agency listing, building code and or electrical code requirements.

4 Dimensional information

4.1 General

The figures provided in this document give normative dimensions essential for interchangeability and performance requirements of this standard but apart from this the drawings are not intended to govern design.

Dimensions are in millimetres except where noted.

Figure 3 illustrates variant 01 drawing 1;

Figure 4 illustrates variant 01 drawing 2;

Figure 5 illustrates variant 02 drawing;

Figure 6 illustrates variant 03 drawing 1;

Figure 7 illustrates variant 03 drawing 2;

Figure 8 illustrates variant 03 drawing 3;

Figure 9 illustrates variant 04 drawing 1;

Figure 10 illustrates variant 04 drawing 2;

Figure 11 illustrates variant 05 drawing 1;

Figure 12 illustrates variant 05 drawing 2;

Figure 13 illustrates variant 05 drawing 3.

4.2 Isometric views and common features

4.2.1 General

Figures 1 and 2 display isometric views fixed connectors and free connectors.

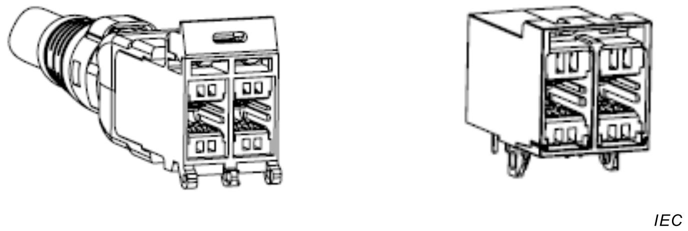


Figure 1 – Isometric view of cable and PCB fixed connectors

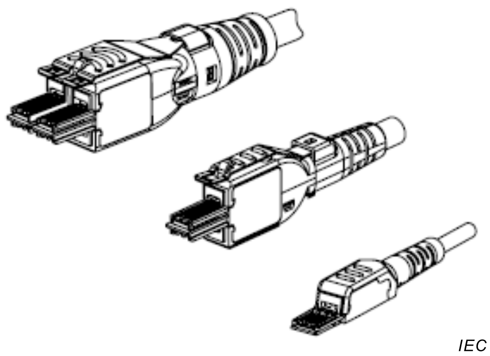


Figure 2 – Isometric view showing free connector for 4, 2 and 1 pair

4.2.2 Fixed connector variant 01 (cable outlet)

Figures 3 and 4 illustrate variant 01 (cable outlet). The corresponding dimensions are shown in Table 1 and Table 2.

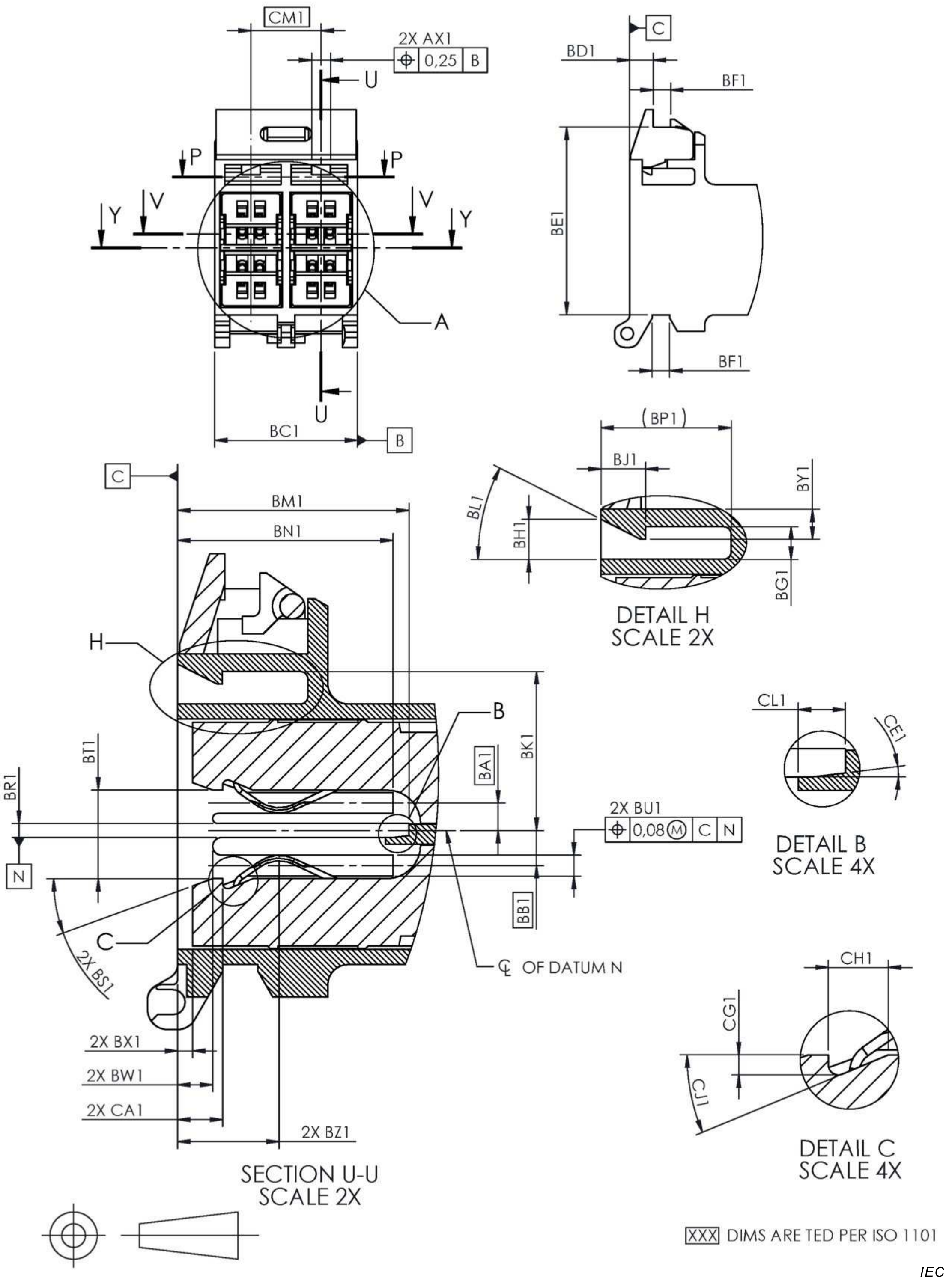


Figure 3 – Variant 01 drawing 1

Table 1 – Variant 01 drawing 1 dimensions

Dimensions in millimetres

Letter	**Minimum**	**Nominal** (ref.)	**Maximum**
AX1	1,83	1,91	1,98
BC1	14,35	14,48	14,61
BD1	Reference	2,39	Reference
BE1	Reference	19,05	Reference
BF1	Reference	1,78	Reference
BG1	1,42	1,65	1,88
BH1	1,88	2,03	2,18
BJ1	2,18	2,26	2,34
BK1	7,91	8,06	8,22
BL1	24°	27°	30°
BM1	11,66	11,73	11,81
BN1	10,82	10,92	11,00
BP1	Reference	6,60	Reference
BR1	0,64	0,71	0,79
BS1	17°	20°	23°
BT1	4,42	4,65	4,85
BU1	0,99	1,07	1,14
BW1	1,68	1,78	1,85
BX1	0,56	0,76	0,99
BY1	0,56	0,64	0,71
BZ1	4,75	4,95	5,15
CA1	2,03	2,31	2,59
CM1	TED	7,11	TED
BA1	TED	1,40	TED
BB1	TED	1,78	TED
CE1	6°	7°	8°
CG1	0,43	0,51	0,58
CH1	1,40	1,52	1,65
CJ1	18°	23°	24°
CL1	1,14	1,19	1,24

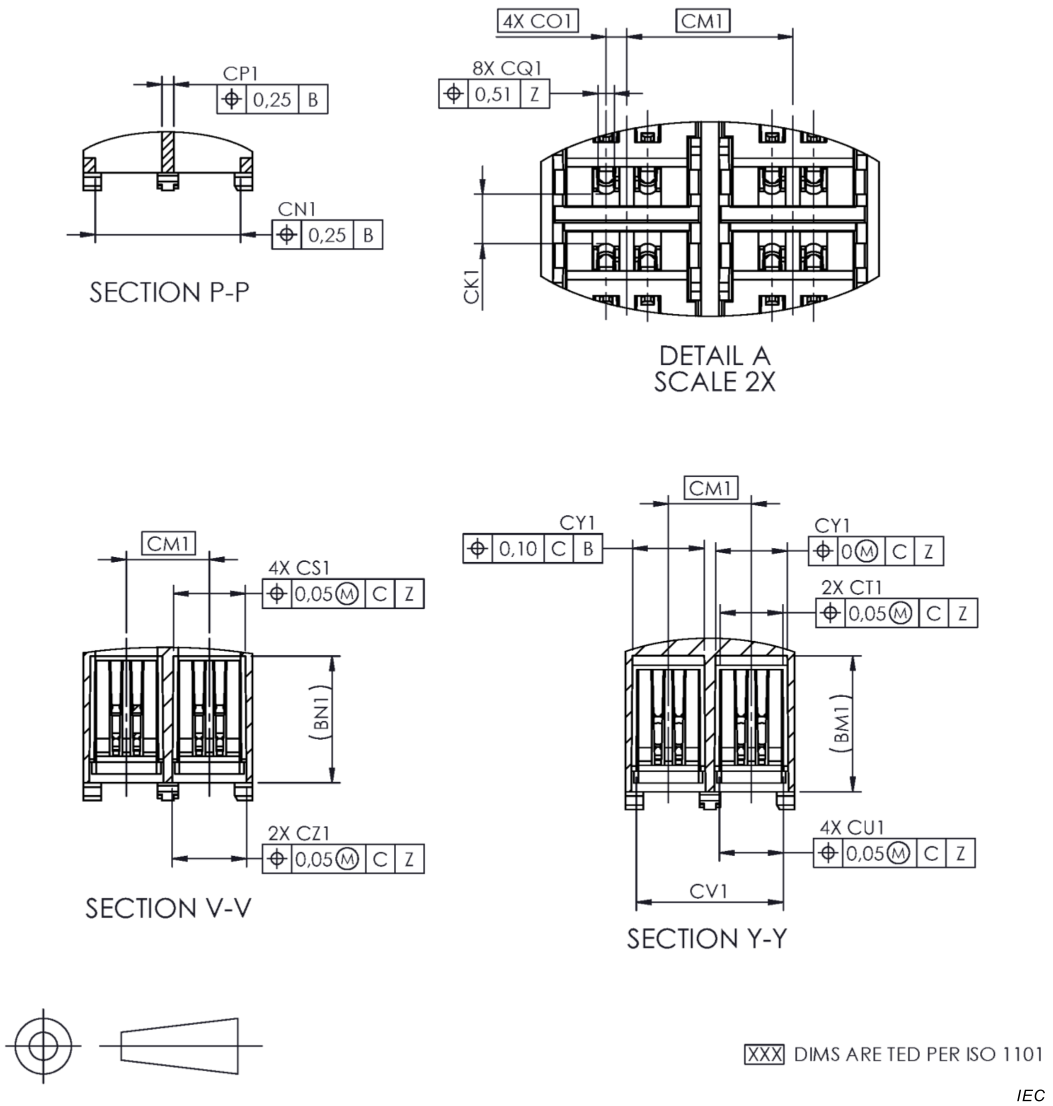


Figure 4 – Variant 01 drawing 2

Table 2 – Variant 01 drawing 2 dimensions

Dimensions in millimetres

Letter	Minimum	Nominal (ref.)	Maximum
CK1	1,80	2,24	2,62
CN1	12,32	12,45	
CP1		1,02	1,14
CS1	6,10	6,17	6,25
CT1	5,28	5,36	5,44
CU1	5,44	5,49	5,54
CV1	12,52	12,60	12,67
CY1	6,07	6,15	6,22
CZ1	6,27	6,35	6,43
CQ1			0,69
CO1	TED	0,89	TED

4.2.3 Fixed connector variant 02, (printed circuit board outlet)

Figure 5 illustrates variant 02 (Printed circuit board outlet). The corresponding dimensions are shown in Table 3.

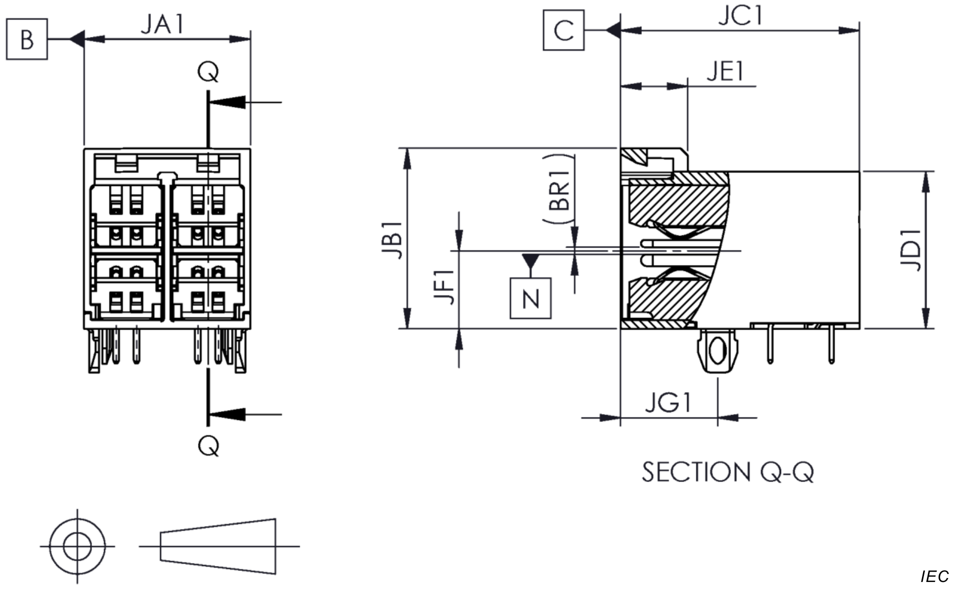


Figure 5 – Variant 02 drawing

Table 3 – Variant 02 drawing dimensions

Dimensions in millimetres

Letter	**Minimum**	**Nominal** (ref.)	**Maximum**
JA1		14,48	14,61
JB1		15,75	16,00
JC1	Reference	20,78	Reference
JD1	Reference	13,72	Reference
JE1	Reference	5,84	Reference
JF1	6,79		
JG1	8,38	8,46	8,53
NOTE See Figures 3 and 4 for internal dimensions.			

4.2.4 Free connector variant 03 (4-pair plug)

Figures 6, 7, and 8 illustrate variant 03 (4-pair plug). The corresponding dimensions are shown in Tables 4, 5 and 6.

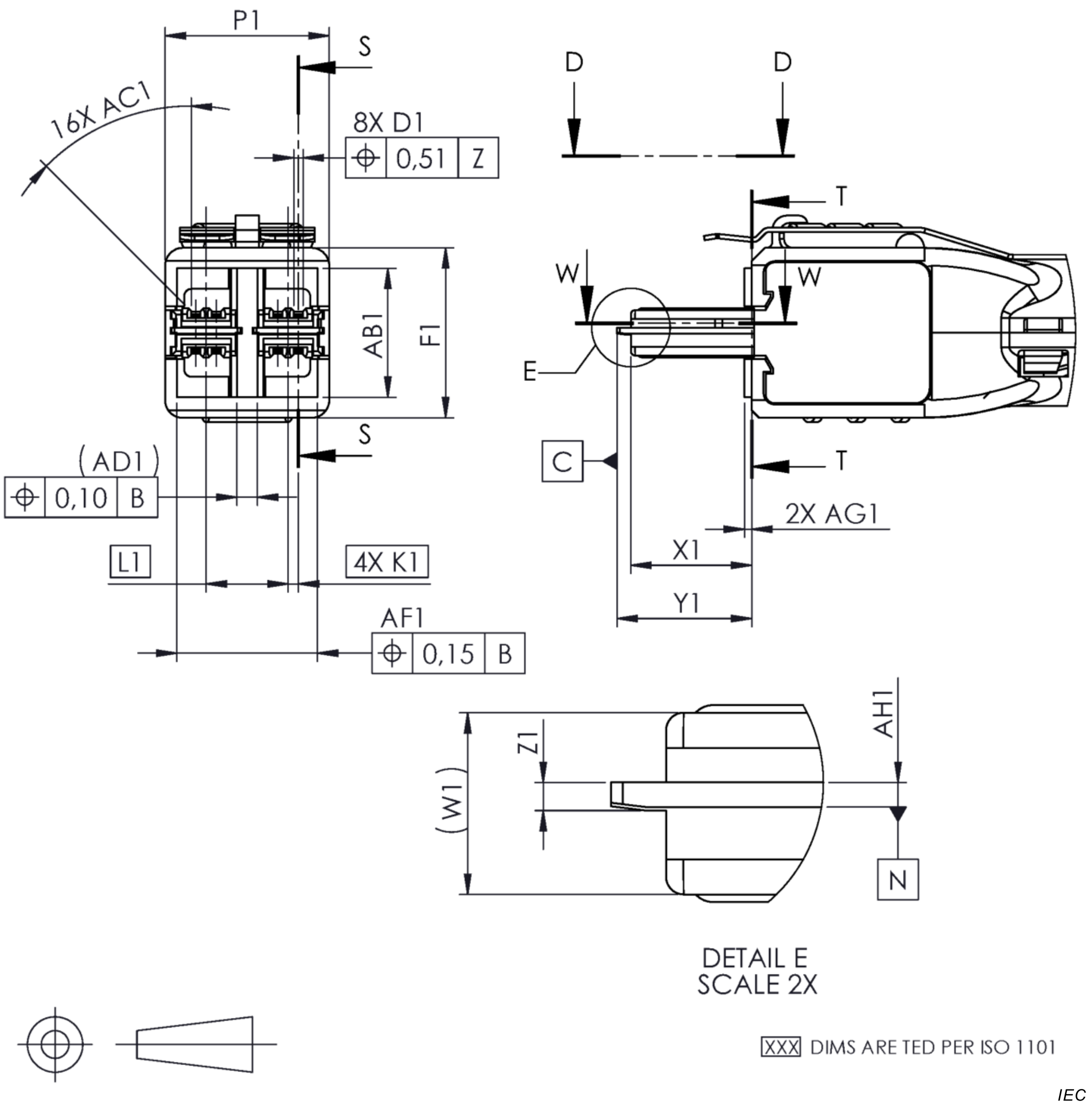


Figure 6 – Variant 03 drawing 1

Table 4 – Variant 03 drawing 1 dimensions

Dimensions in millimetres

Letter	Minimum	Nominal (ref.)	Maximum
D1[a]	0,76	0,81	0,86
F1		14,87	
K1	TED	0,89	TED
L1	TED	7,11	TED
P1		14,22	14,38
W1		3,84 Reference	
X1	10,28	10,46	10,64
Y1	11,40	11,66	11,91
Z1	0,58	0,61	0,64
AB1		11,3	11,53
AC1	44° x 0,13	45° x 0,25	46° x 0,38
AD1		1,78 Reference	
AF1		12,26	12,39
AG1	0,56	0,64	0,71
AH1	0,48	0,53	0,58

[a] Dimension D1 is the same as ES1 in Figure 10 and GS1 in Figure 13

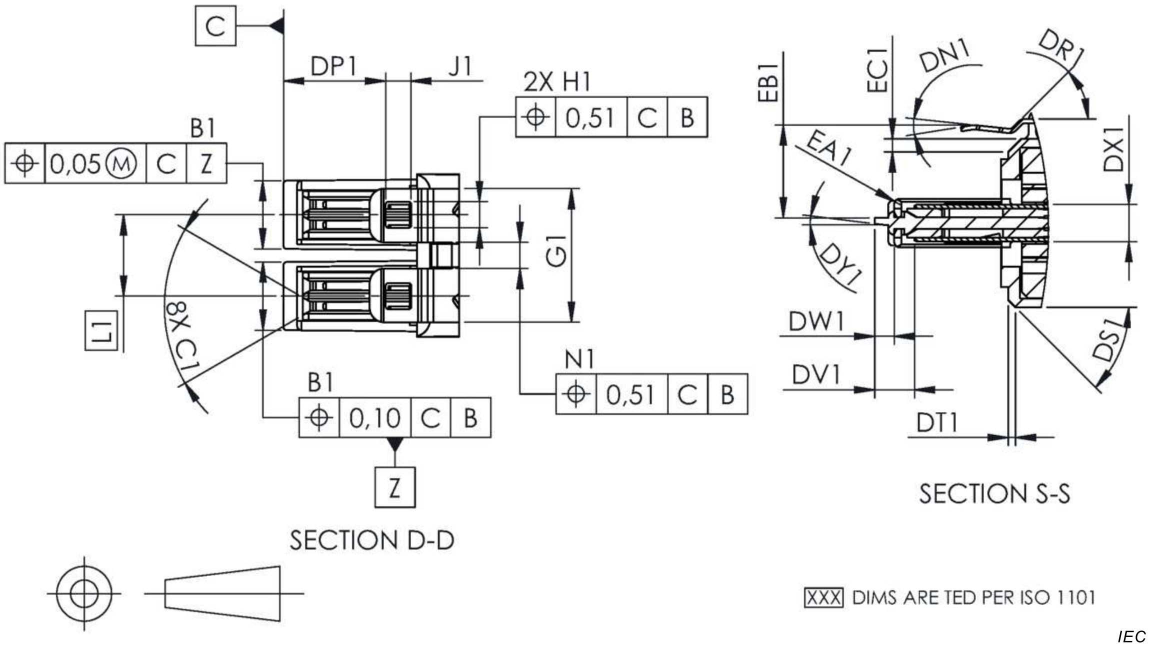


Figure 7 – Variant 03 drawing 2

Table 5 – Variant 03 drawing 2 dimensions

Dimensions in millimetres

Letter	**Minimum**	**Nominal** (ref.)	**Maximum**
B1	5,92	5,97	6,02
C1	58°	60°	62°
G1	11,56	11,68	11,81
H1	2,21	2,29	
J1	2,13	2,26	2,39
N1	2,21	2,29	2,36
DN1	12°	15°	18°
DP1	8,61	9,04	9,47
DR1	Reference	45°	Reference
DS1	44°	45°	46°
DT1	0,51	0,64	0,76
DV1	3,20	3,48	3,86
DW1		0,64	0,76
DX1	3,02	3,25	3,48
DY1	6°	7°	8°
EA1	R 0,38	R 0,51	R 0,66
EB1	7,76	8,10	8,56
EC1	1,02	1,14	1,27

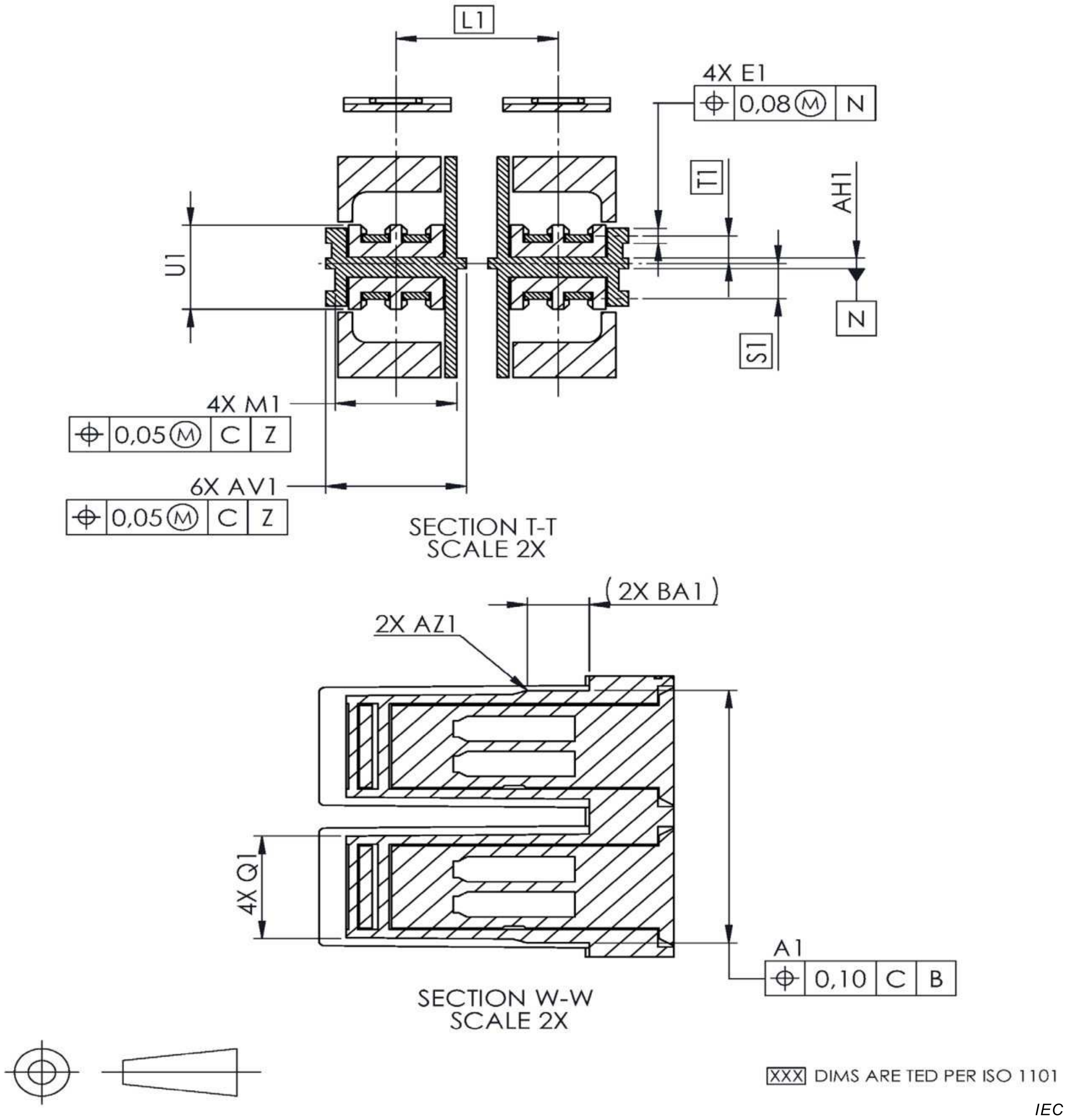


Figure 8 – Variant 03 drawing 3

Table 6 – Variant 03 drawing 3 dimensions

Dimensions in millimetres

Letter	**Minimum**	**Nominal** (ref.)	**Maximum**
A1	12,73	12,80	12,85
E1	0,69	0,76	0,84
M1	5,26	5,33	5,41
Q1	5,08	5,13	5,18
S1	TED	1,78	TED
T1	TED	1,40	TED
U1	4,12	4,27	4,42
AV1	6,10	6,20	6,25
AZ1	R 1,96	R 2,03	R 2,11
BA1	Reference	2,72	Reference

4.2.5 Free connector variant 04 (2-pair plug)

Figures 9 and 10 illustrate variant 04 (2-pair plug). The corresponding dimensions are shown in Tables 7 and 8.

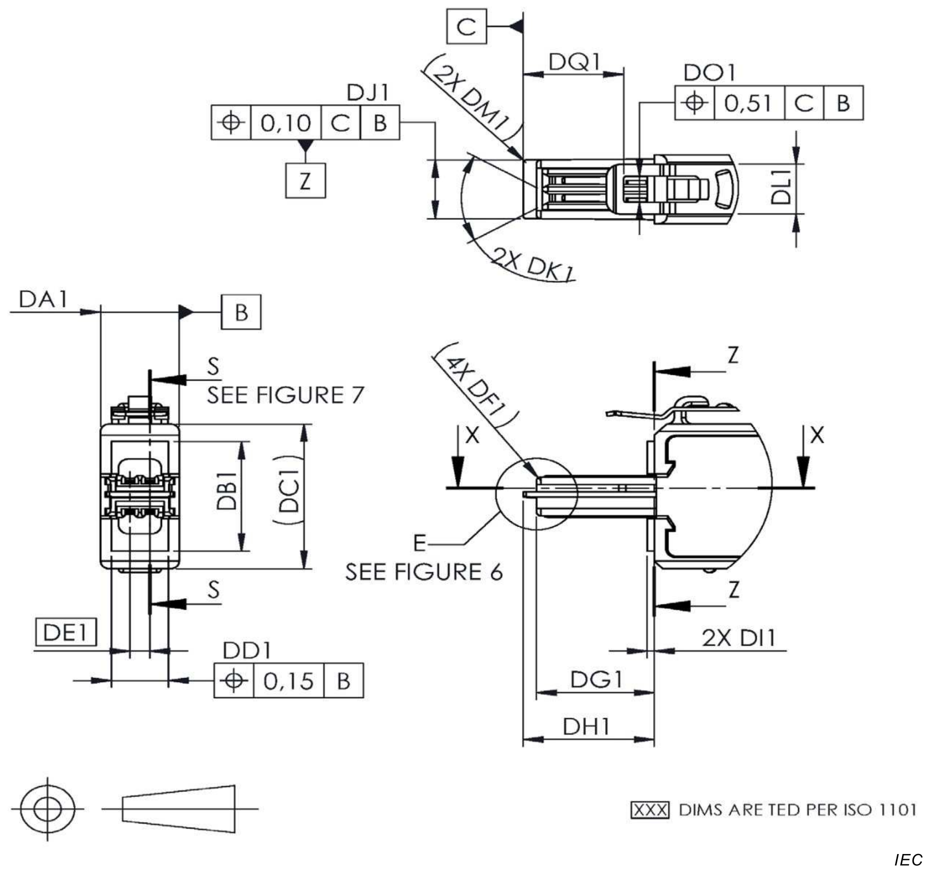


Figure 9 – Variant 04 drawing 1

Table 7 – Variant 04 drawing 1 dimensions

Dimensions in millimetres

Letter	**Minimum**	**Nominal** (ref.)	**Maximum**
DA1		7,01	7,16
DB1		11,30	11,53
DC1	Reference	14,87	Reference
DD1		5,08	5,28
DE1	TED	1,78	TED
DF1	Reference	0,38	Reference
DG1	10,28	10,46	10,64
DH1	11,40	11,66	11,91
DI1	0,56	0,64	0,71
DJ1	5,92	5,97	6,02
DK1	58°	60°	62°
DL1	4,95	5,08	5,21
DM1	Reference	0,25	Reference
DO1	2,21	2,29	
DQ1	8,67	9,04	9,47

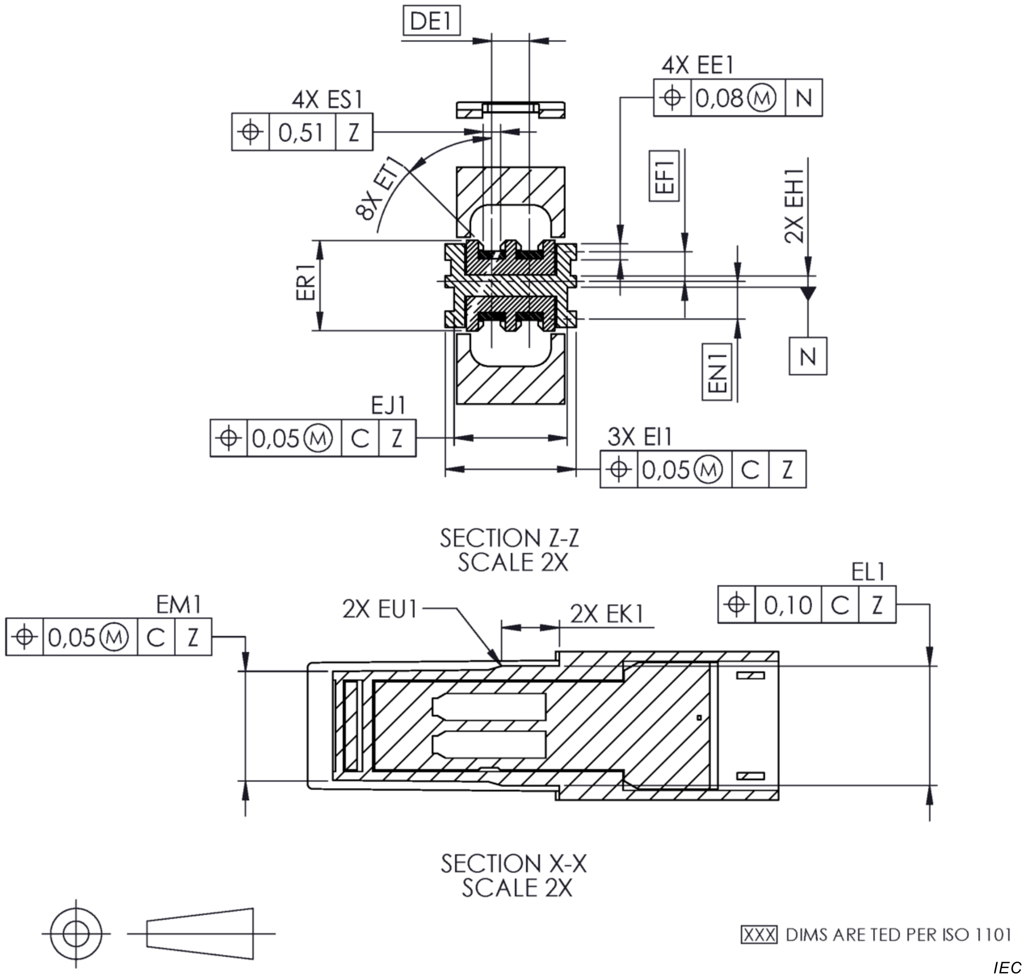


Figure 10 – Variant 04 drawing 2

Table 8 – Variant 04 drawing 2 dimensions

Dimensions in millimetres

Letter	Minimum	Nominal (ref.)	Maximum
EE1	0,69	0,76	0,84
EF1	TED	1,40	TED
EH1	0,48	0,53	0,58
EI1	6,10	6,17	6,25
EJ1	5,13	5,33	5,41
EK1		2,72 Reference	
EL1	5,61	5,64	5,69
EM1	5,08	5,13	5,18
EN1	TED	1,78	TED
ER1	4,12	4,27	4,42
ES1 [a]	0,76	0,81	0,86
ET1	44°X0,13	45°X0,25	46°X0,38
EU1	R 1,96	R 2,03	R 2,11

[a] Dimension ES1 is the same as D1 in Figure 6 and GS1 in Figure 13.

4.2.6 Free connector variant 05 (1-pair plug)

Figures 11, 12, and 13 illustrate variant 05 (1-pair plug). The corresponding dimensions are shown in Tables 9, 10 and 11.

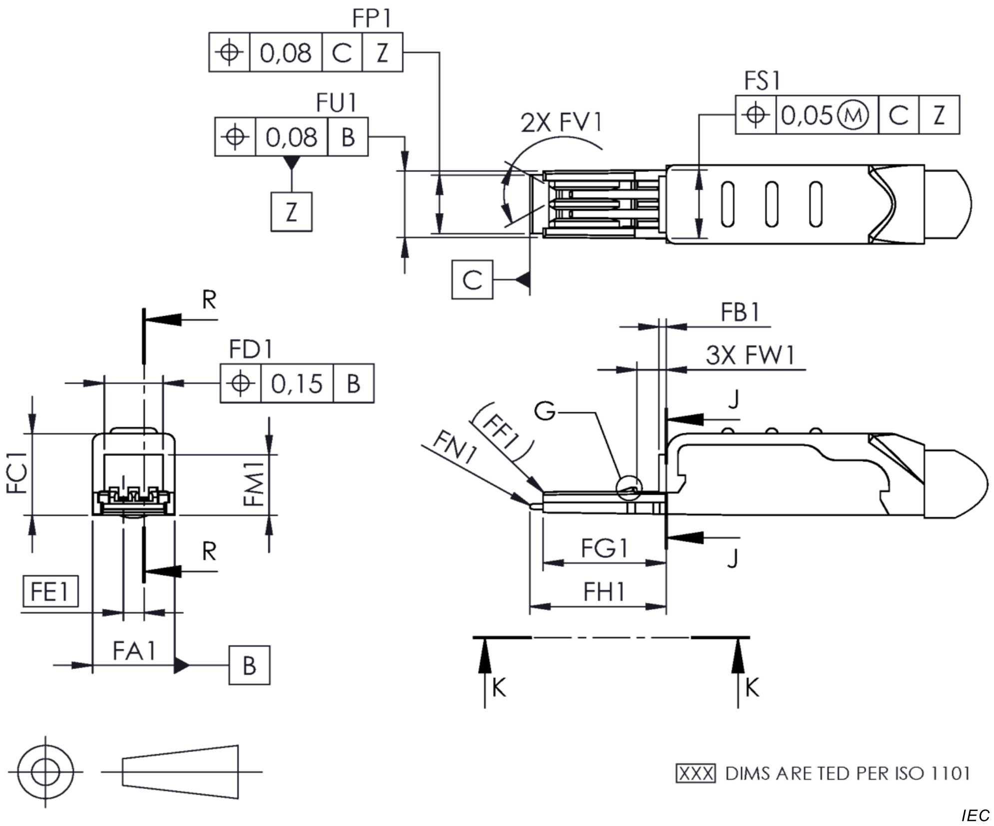


Figure 11 – Variant 05 drawing 1

Table 9 – Variant 05 drawing 1 dimensions

Dimensions in millimetres

Letter	**Minimum**	**Nominal** (ref.)	**Maximum**
FA1		7,11	7,19
FB1	0,56	0,64	0,71
FC1	7,29	7,37	7,52
FD1	4,95	5,08	5,28
FE1	TED	1,78	TED
FF1	Reference	R 0,25	Reference
FG1	10,62	10,64	10,67
FH1	11,71	11,76	11,81
FM1		5,54	5,76
FN1	R 0,15	R 0,23	R 0,31
FP1	5,23	5,28	5,33
FS1	6,10	6,20	6,25
FU1	5,97	6,02	6,07
FV1	58°	60°	62°
FW1	2,24	2,36	2,49

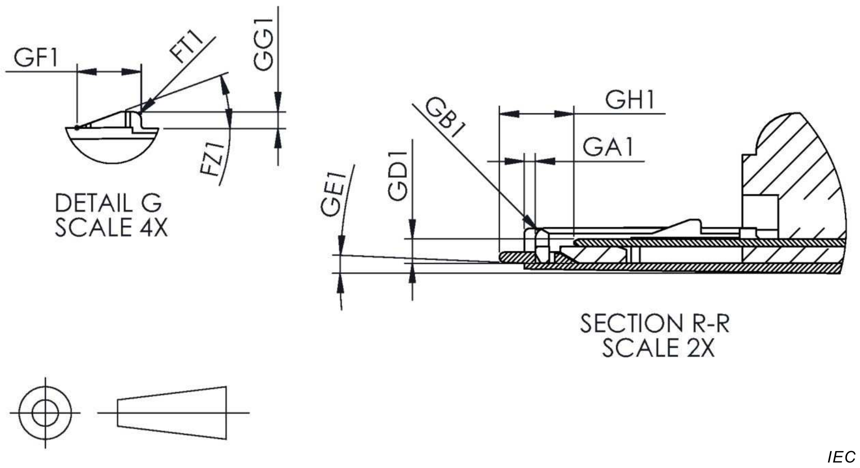


Figure 12 – Variant 05 drawing 2

Table 10 – Variant 05 drawing 2 dimensions

Dimensions in millimetres

Letter	**Minimum**	**Nominal** (ref.)	**Maximum**
FT1	R 0,05	R 0,18	R 0,30
FZ1	19°	20°	21°
GA1	0,56	0,66	0,76
GB1	0,38	0,51	0,64
GD1	1,47	1,63	1,78
GE1	1,5°	2,5°	3,5°
GF1	1,40	1,52	1,65
GG1	0,30	0,38	0,46
GH1	3,20	3,48	3,76

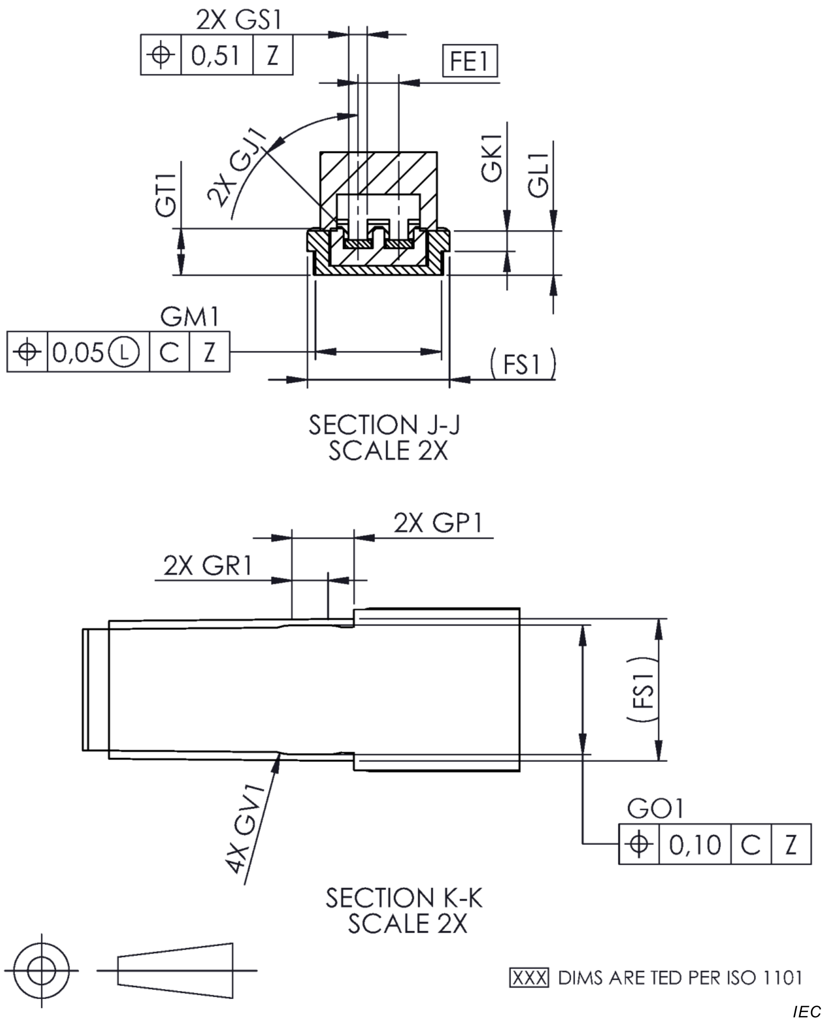


Figure 13 – Variant 05 drawing 3

Table 11 – Variant 05 drawing 3 dimensions

Dimensions in millimetres

Letter	Minimum	Nominal (ref.)	Maximum
GJ1	44°x 0,13	45°x 0,25	46°x 0,38
GK1	0,80	0,89	0,98
GL1	2,01	2,11	2,21
GM1	5,41	5,49	5,56
GO1	5,61	5,64	5,66
GP1	2,64	2,72	2,79
GR1	1,19	1,37	
GS1 [a]	0,76	0,81	0,86
GT1	1,93	2,13	2,34
GV1	R 1,96	R 2,03	R 2,11

[a] Dimension GS1 is the same as D1 in Figure 6 and ES1 in Figure 10.

4.3 Terminations

4.3.1 General

A connector may include multiple terminations between the cable termination and the separable contact interface. These may include press-in (compliant pin) connections of jack springs into PCBs, for example. All terminations shall meet the relevant termination requirements.

If a type of solderless termination is used which is not covered by any IEC 60352 series specification, the supplier shall assure its reliability by performing similar tests, and demonstrating a similar level of performance, to those in IEC 60352. The supplier shall hold test results demonstrating reliability and performance and make such copies of such results available upon request.

Free connectors are intended to be terminated to cable to provide connector and cable assemblies. The connector type designation provides basic terminations concerning the type of conductor (tinsel, stranded, solid) to which the conductor may be applied, and the type of connection used (solder, insulation displacement, etc.).

NOTE Specific details concerning wire gauge size, type and thickness of conductor insulation, size and shape of cordage or cable jacket, etc., are not intended to be part of this specification. Minor variations in a free connector's interior details to accommodate differing wire gauge sizes, outer jackets, etc., do not require the generation of new free connector specifications.

If the connector is able to be used with multiple cable types, the manufacturer shall assure that all connection points conform with the relevant IEC standards with each of the cable types.

4.3.2 Referenced solderless termination types

4.3.2.1 General

If a type of solderless termination is used, which is covered by an IEC 60352 series specification, the supplier shall assure its reliability by performing tests and demonstrating a level of performance, according to the IEC 60352 series. The supplier shall hold test results demonstrating reliability and performance and make such copies of such results available upon request.

4.3.2.2 Insulation displacement terminations

Insulation displacement terminations shall conform to IEC 60352-3 or IEC 60352-4.

4.3.2.3 Crimp terminations

Crimp terminations shall conform to IEC 60352-2.

4.3.2.4 Insulation piercing terminations

Insulation piercing terminations shall conform to IEC 60352-6.

4.3.2.5 Compliant pin (press-in)

Compliant pin (press-in) shall conform to IEC 60352-5.

4.3.2.6 Spring clamp terminations

Spring clamp terminations shall conform to IEC 60352-7.

4.3.3 Non-referenced termination types

If a type of termination is used which is not covered by any IEC 60352 series specification, the supplier shall assure its reliability by performing similar tests, and demonstrating a similar level of performance, to those covered by IEC 60352 series. The supplier shall hold test results demonstrating reliability and performance and provide copies of such results available upon request.

4.3.4 Solder terminations

Solder terminations are not covered in the IEC 60352 series and are allowed.

4.4 Mounting information

4.4.1 General

To be determined by the manufacturer.

4.4.2 Mounting information for fixed connectors

4.4.2.1 General

Fixed mounting information shall be specified by the manufacturer for the relevant connector.

4.4.2.2 Hole pattern on printed boards

Fixed connector printed board mounting shall be specified by the manufacturer for the relevant connector.

4.4.2.3 Mounting on panels

Fixed connector mounting on panels shall be specified by the manufacturer for the relevant connector.

4.4.3 Mounting information for free connectors

Mounting information for free connectors onto cables shall be specified by the manufacturer for the relevant connector.

4.5 Gauges

4.5.1 Sizing gauges and retention force gauges; mechanical function, engaging/separating/insertion/withdrawal force

4.5.1.1 Fixed connector (outlet) gauges

Gauges shall be made according to the following requirements:

– material: tool steel, hardened.

– $\sqrt{}$ = Surface roughness, according to ISO 1302;

– Ra = 0,25 μm max.;

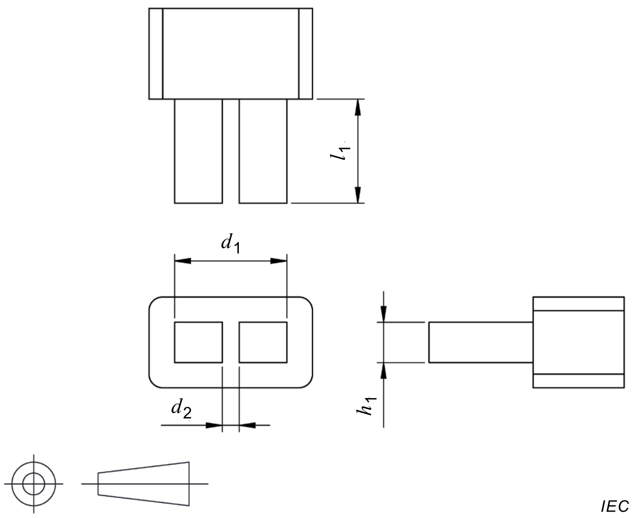
– A 0,01 mm wear tolerance may be applied, keeping the tolerance always on the safe side (the relevant GO or NO-GO test shall never be facilitated by the foreseeable gauge wear);

– clearance shall be provided for connector contacts.

NOTE Annex A provides additional information about using the gauges.

See Figures 14 to 17 below for illustrations and dimensions of "Go" and "No-Go" gauges for fixed connectors.

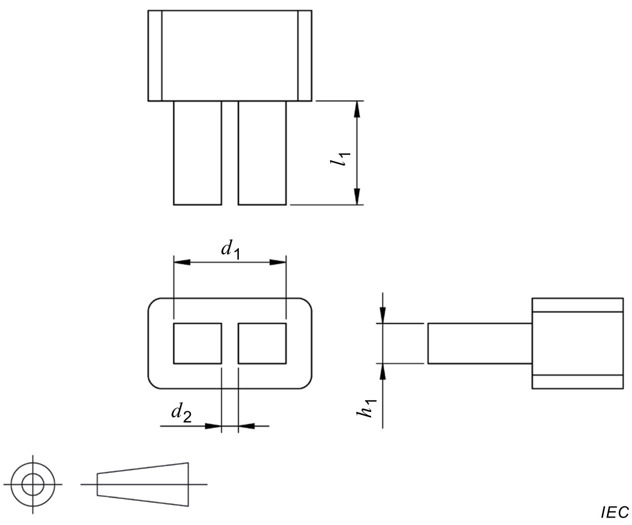
Dimensions in millimetres



Letter	Minimum	Nominal (ref.)	Maximum
d_1	12,38	12,37	12,36
d_2	1,84	1,85	1,86
h_1	4,42	4,43	4,44
l_1	11,42	11,47	11,52

Figure 14 – Fixed connector location "Go" gauge

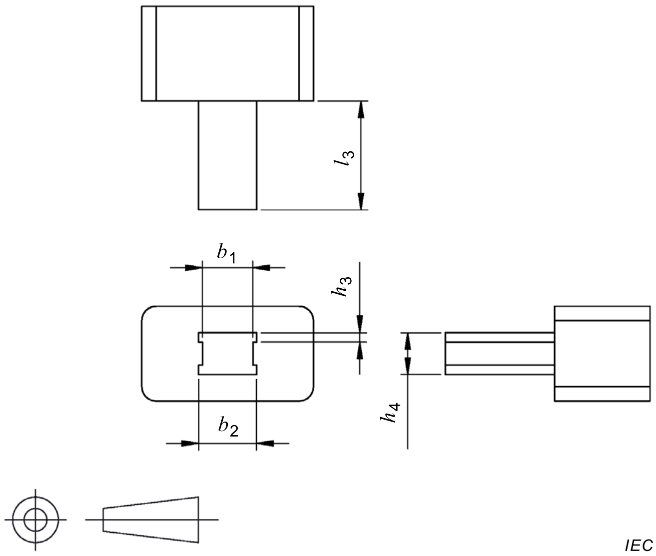
Dimensions in millimetres



Letter	Minimum	Nominal (ref.)	Maximum
d_3	12,71	12,72	12,73
d_4	1,49	1,50	1,51
h_2	4,83	4,84	4,85
l_2	11,42	11,47	11,52

Figure 15 – Fixed connector location "No-Go" gauge

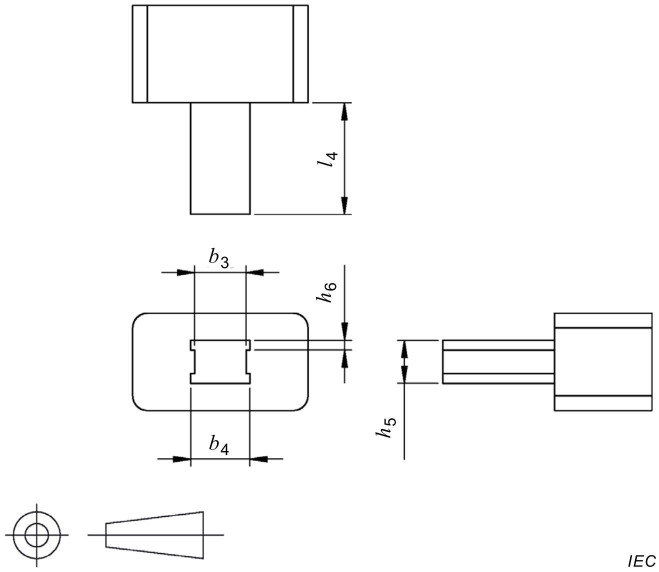
Dimensions in millimetres



Letter	Minimum	Nominal (ref.)	Maximum
b_1	5,28	5,29	5,30
b_2	6,07	6,08	6,09
h_3	0,99	1,00	1,01
h_4	4,42	4,43	4,44
l_3	11,42	11,47	11,52

Figure 16 – Fixed connector size "Go" gauge

Dimensions in millimetres



Letter	Minimum	Nominal (ref.)	Maximum
b_3	5,52	5,53	5,54
b_4	6,41	6,42	6,43
h_5	4,83	4,84	4,85
h_6	1,12	1,13	1,14
l_4	11,42	11,47	11,52

Figure 17 – Fixed connector size "No-Go" gauge

4.5.1.2 Free connector (plug) gauges

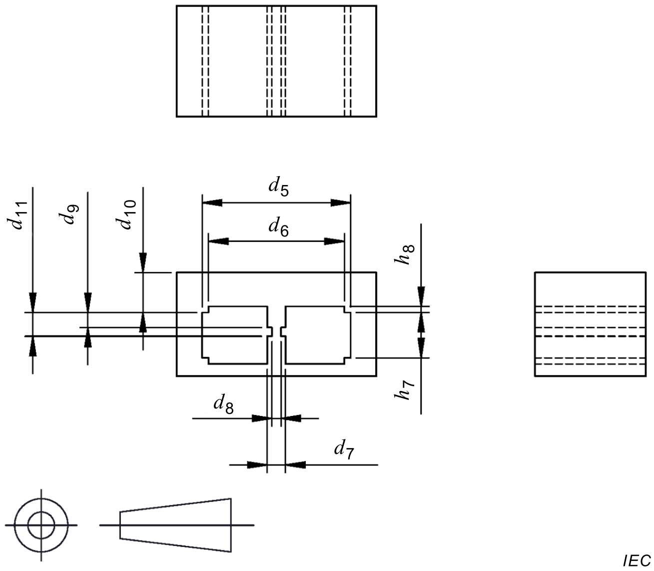
Gauges shall be made according to the following requirements:

- material: tool steel, hardened;
- $\bigtriangledown\!\!/$ = Surface roughness, according to ISO 1302;
- Ra = 0,25 μm max.;
- A 0,01 mm wear tolerance may be applied, keeping the tolerance always on the safe side (the relevant GO or NO-GO test shall never be facilitated by the foreseeable gauge wear);
- clearance shall be provided for connector contacts.

NOTE Annex A provides additional information about using the gauges.

See Figures 18 to 21 below for illustrations and dimensions of "Go" and "No-Go" gauges for free connectors.

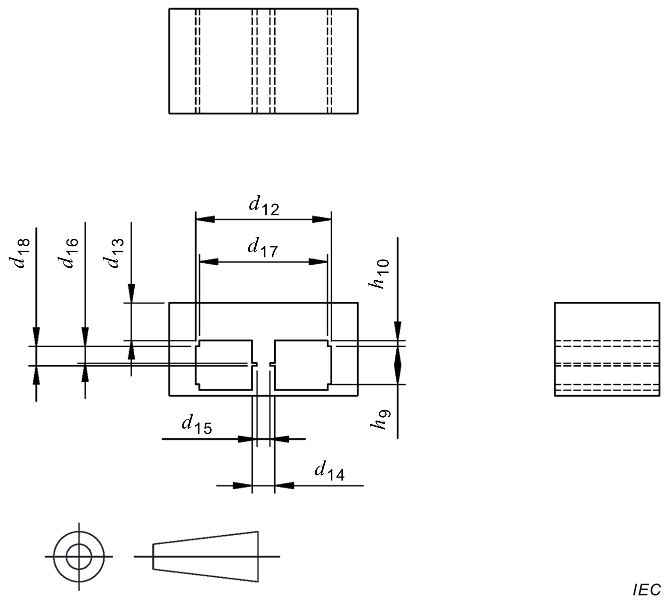
Dimensions in millimetres



Letter	Minimum	Nominal (ref.)	Maximum
d_5	13,37	13,38	13,39
d_6	3,58	3,60	3,62
d_7	1,64	1,65	1,66
d_8	0,83	0,84	0,85
d_9	1,34	1,35	1,36
d_{10}	12,24	1,26	12,28
d_{11}	2,13	2,14	2,15
h_7	4,08	4,09	4,10
h_8	0,53	0,55	0,57

Figure 18 – Free connector location "Go" gauge

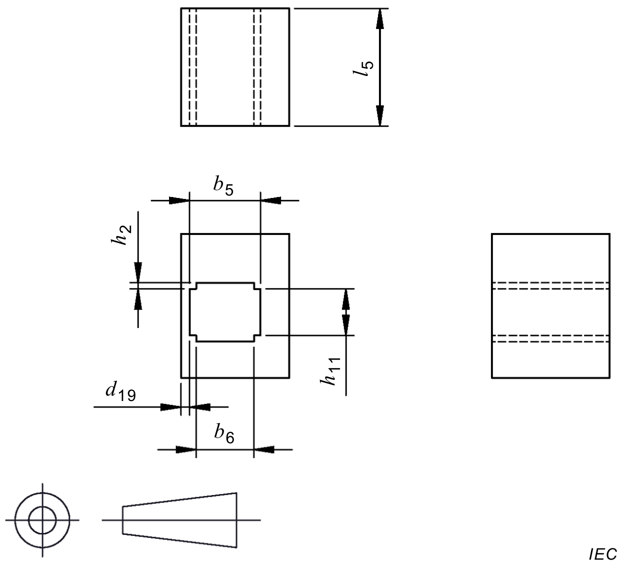
Dimensions in millimetres



Letter	Minimum	Nominal (ref.)	Maximum
d_{12}	12,95	12,96	12,97
d_{13}	3,58	3,60	3,62
d_{14}	2,17	2,18	2,19
d_{15}	1,25	1,26	1,27
d_{16}	1,60	1,61	1,62
d_{17}	12,24	12,26	12,28
d_{18}	1,87	1,88	1,89
h_9	3,64	3,65	3,66
h_{10}	0,53	0,55	0,57

Figure 19 – Free connector location "No-Go" gauge

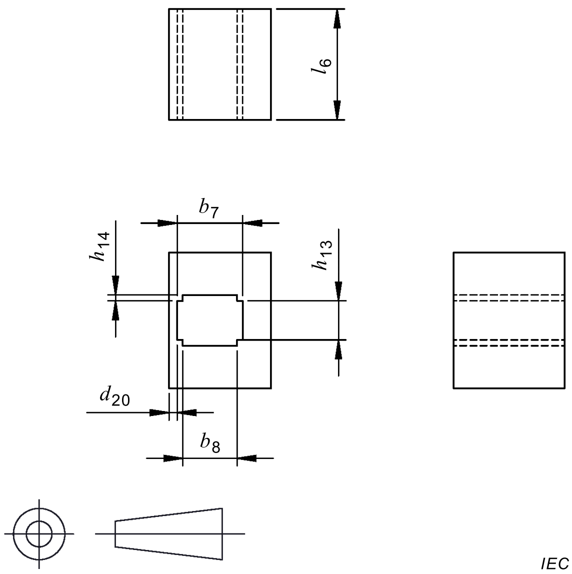
Dimensions in millimetres



Letter	Minimum	Nominal (ref.)	Maximum
b_5	6,23	6,24	6,25
b_6	5,06	5,08	5,10
d_{19}	0,74	0,76	0,78
h_{11}	4,08	4,09	4,10
h_{12}	0,53	0,55	0,57
l_5	10,32	10,37	10,42

Figure 20 – Free connector size "Go" gauge

Dimensions in millimetres



Letter	Minimum	Nominal (ref.)	Maximum
b_7	6,10	6,11	6,12
b_8	5,06	5,08	5,10
d_{20}	0,74	0,76	0,78
h_{13}	3,64	3,65	3,66
h_{14}	0,53	0,55	0,57
l_6	10,32	10,37	10,42

Figure 21 – Free connector size "No-Go" gauge

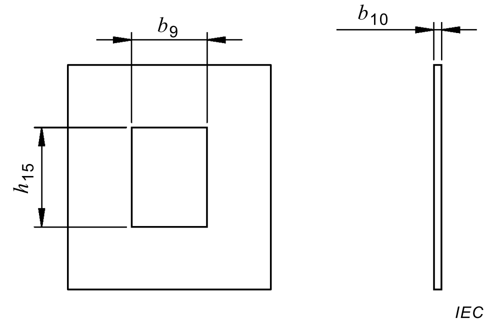
4.6 Mechanical function, engaging/separating/insertion/withdrawal force gauges

Mechanical function shall be determined using the sizing gauges given in 4.5.1 so as to meet requirements given in 4.5.1.

4.7 Test panels

Test panels for panel mounted fixed connectors shall be as defined in Figure 22.

Dimensions in millimetres



Letter	Minimum	**Nominal** (ref.)	Maximum
b_9	14,61	14,68	14,91
b_{10}	1,50	1,57	1,65
h_{15}	19,23	19,30	19,38

Figure 22 – Fixed connector panel

5 Characteristics

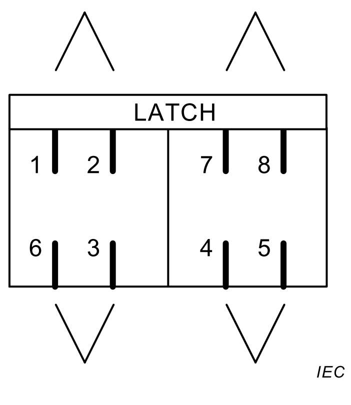
5.1 General

Conformance to the test schedules shall ensure the reliability of all performance parameters, including transmission parameters.

NOTE Stable and conforming contact resistance is a good indication of the stability of transmission performance.

5.2 Pin and pair grouping assignment

For those specifications where pin and pair groupings are relevant, the pin and pair grouping assignments shall be as shown in Figure 23, unless otherwise specified.



NOTE The pin designations correspond to those of IEC 60603-7 series interface.

Figure 23 – Schematic diagram of fixed connector

5.3 Climatic categories

The connectors shall be classified into climatic categories in accordance with the general rules given in IEC 60068-1. The temperature range and severity of the damp heat steady state test shall be as given in Table 12.

NOTE The lower and upper temperatures and the duration of the damp heat, steady state test where selected from the preferred values stated in 2.2 of IEC 61076-1:2006.

Table 12 – Climatic categories – Selected values

Climatic category	Lower temperature °C	Upper temperature °C	Damp heat steady state days
40/070/21	–40	70	21

5.4 Electrical characteristics

5.4.1 Creepage and clearance distances

NOTE 1 The permissible operating voltages depend on the application and on the applicable or specified safety requirements.

NOTE 2 Insulation co-ordination is not required for this connector; therefore, the creepage and clearance distances in 2.4 of IEC 61076-1:2006 are reduced. This performance aspect is taken into account in the overall performance requirements.

The creepage and clearance distances shall be representative of the operating characteristics of mated connectors and shall be verified as complying with Table 13.

In practice, reductions in creepage or clearance distances may occur due to the conductive pattern of the printed board or the wiring used, and shall duly be taken into account.

Table 13 – minimum creepage and clearance distances

Dimensions in millimetres

Minimum distance between contacts and chassis		Minimum distance between adjacent contacts	
Creepage	Clearance	Creepage	Clearance
1,40	0,51	0,36	0,36

5.4.2 Voltage proof

Conditions: IEC 60512-4-1, Test 4a, Method A

Mated connectors: standard atmospheric conditions

All variants:

- 1 000 V DC or AC peak, contact-to-contact
- 1 500 V DC or AC peak, contact-to-test panel or shield

5.4.3 Current carrying capacity

Conditions: IEC 60512-5-2, Test 5b

All contacts, connected in series

The current-carrying capacity of connectors in accordance with the requirements of 2.5 of IEC 61076-1:2006 shall conform with the de-rating curve given in Figure 24.

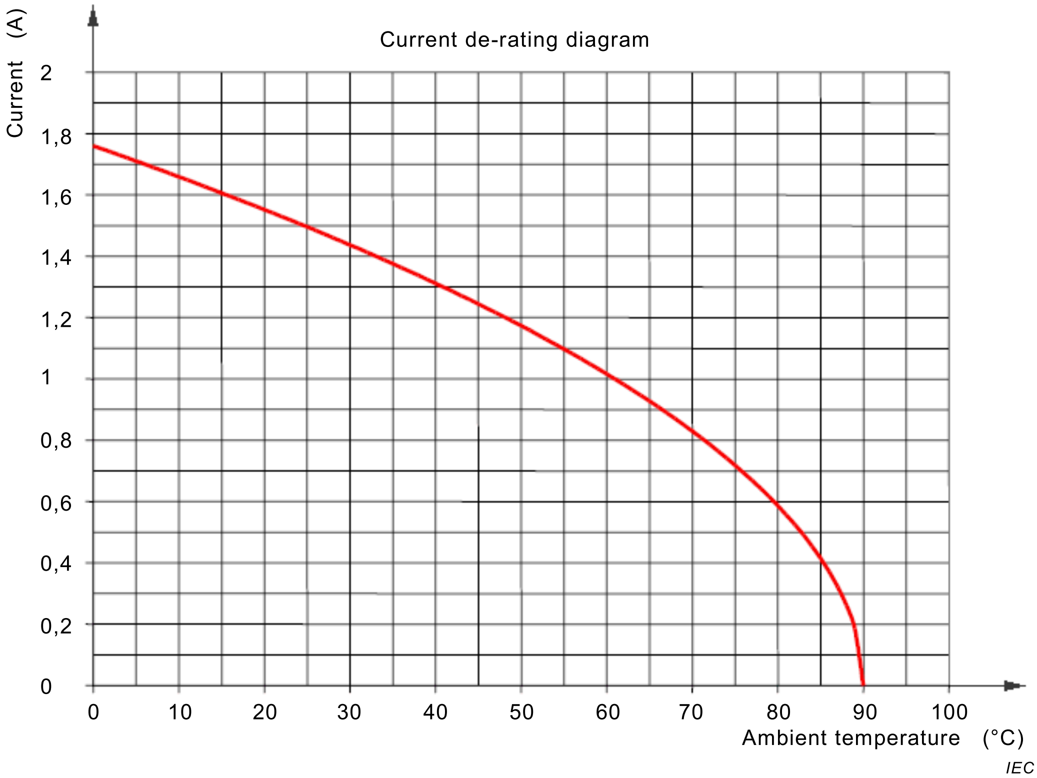


Figure 24 – Connector de-rating curve

5.4.4 Initial contact and shield resistance

Conditions: IEC 60512-2-1, test 2a

Mated connectors

Signal contacts: 20 mΩ maximum.

Shield Contacts: 20 mΩ maximum.

5.4.5 Input to output resistance

Conditions: IEC 60512-2-1, test 2a

Mated connectors

Signal contacts: 200 mΩ maximum.

Shield contacts: 100 mΩ maximum.

5.4.6 Input to output resistance unbalance

Conditions: IEC 60512-2-1, test 2a

Mated connectors

Among all signal conductors, maximum difference between maximum and minimum

50 mΩ maximum

5.4.7 Insulation resistance

Conditions: IEC 60512-3-1, test 3a, Method A

Mated connectors

Test voltage: 100 V direct current

Each contact and screen to all others: 500 MΩ minimum

5.5 Transmission characteristics

5.5.1 General

The performance levels, respective to transmission characteristics, are determined according to specific test methods given in 5.5.2 to 5.5.14.

The transmission performance interoperability of independent fixed or free connectors is determined by tests conducted when mated with precision test fixtures.

All transmission performance requirements shall apply between the reference planes specified in Clause D.8.

5.5.2 Insertion loss

Conditions: Mated connectors, all pairs of contacts. See Table 14 below.

Table 14 – Insertion loss

Frequency MHz	**Maximum insertion loss** [a] dB		**Test method**
	Connector category		
	Category 7_A [b]	**Category 8.2**	
$1 \leq f \leq 1\,000$	$0{,}02\sqrt{f}$	$0{,}02\sqrt{f}$	IEC 60512-28-100 test 28a
$1\,000 \leq f \leq 2\,000$		$0{,}02\sqrt{(f)} + 0{,}000\,5(f - 1\,000)$	

[a] Insertion loss values that correspond to calculated values of less than 0,1 dB shall revert to a requirement of 0,1 dB.

[b] For alternative measurement methods, see normative Annex E.

5.5.3 Return loss

Conditions: Mated connectors, all pairs of contacts. See Table 15 below.

Table 15 – Return loss

Frequency MHz	Minimum return loss [a] dB Connector category		Test method
	Category 7_A [c]	**Category 8.2**	
$1 \leq f \leq 1\,000$	$68 - 20\ \lg(f)$		IEC 60512-28-100 test 28b
$1 \leq f \leq 2\,000$		$72 - 20\ \lg(f)$[b]	

[a] Return loss values that correspond to calculated values of greater than 30,0 dB shall revert to a minimum requirement of 30,0 dB.

[b] Calculated values below 12,0 dB revert to a 12,0 dB plateau.

[c] For alternative measurement methods, see normative Annex F.

5.5.4 Propagation delay

Conditions: Mated connectors, all pairs of contacts

All types: ≤2,5 ns

NOTE This characteristic does not need to be measured since it is achieved by design.

5.5.5 Delay skew

Conditions: Mated connectors, all pairs of contacts

All types: ≤1,25 ns

NOTE This characteristic does not need to be measured, since it is achieved by design.

5.5.6 NEXT loss

Conditions: Mated connectors, between all combinations of two pairs of contacts. See Table 16 below.

Table 16 – NEXT loss

Frequency MHz	Minimum NEXT loss dB Connector category		Test method
	Category 7_A [a, c]	**Category 8.2 [b]**	
$1 \leq f \leq 600$	$116{,}3 - 20\ \lg(f)$	$116{,}3 - 20\ \lg(f)$	IEC 60512-28-100 test 28c
$600 \leq f \leq 1\,000$	$102{,}4 - 15\ \lg(f)$	$102{,}4 - 15\ \lg(f)$	
$1\,000 \leq f \leq 2\,000$		$57{,}4 - 40\ \lg(f/1\,000)$	

[a] Minimum NEXT loss values that correspond to calculated values of greater than 75,0 dB shall revert to a minimum requirement of 75,0 dB.

[b] Minimum NEXT loss values for Category 8.2 at frequencies that correspond to calculated values of greater than 80,0 dB shall revert to a minimum requirement of 80,0 dB.

[c] For alternative measurement methods, see normative Annex G.

5.5.7 Power sum NEXT loss (for information only)

Conditions: Mated connectors, each pair and all other pairs combined. See Table 17 below.

Table 17 – Power sum NEXT loss

Frequency MHz	Minimum PS NEXT loss [a] dB Connector category		Test method
	Category 7_A [b]	Category 8.2	
$1 \leq f \leq 600$	113,3 – 20 lg(f)	113,3 – 20 lg(f)	IEC 60512-28-100 test 28c
$600 \leq f \leq 1\,000$	99,4 – 15 lg(f)	99,4 – 15 lg(f)	
$1\,000 \leq f \leq 2\,000$		54,4-40 lg(f / 1000)	

[a] PS NEXT loss values that correspond to calculated values of greater than 72,0 dB shall revert to a minimum requirement of 72,0 dB.

[b] For alternative measurement methods see normative Annex G.

NOTE This characteristic is achieved by compliance to NEXT (5.5.6) and there is no necessity to test.

5.5.8 FEXT loss

Conditions: Mated connectors, between all combinations of two pairs of contacts. See Table 18 below.

Table 18 – FEXT loss

Frequency MHz	Minimum FEXT loss [c] dB Connector category		Test method
	Category 7_A [a, d]	Category 8.2 [b]	
$1 \leq f \leq 1\,000$	90 – 15 lg(f)	103,9 – 20 lg10(f)	IEC 60512-28-100 test 28d
$1\,000 \leq f \leq 2\,000$		43,9 – 60 lg10(f /1 000)	

[a] For IEC 61076-3-104 connectors. FEXT loss values that correspond to calculated values of greater than 75,0 dB shall revert to a minimum requirement of 75,0 dB.

[b] For IEC 61076-3-104 Catergory 8.2 connectors. FEXT loss values that correspond to calculated values of greater than 80,0 dB shall revert to a minimum requirement of 80,0 dB.

[c] For connectors, the difference between FEXT loss and ACR-F is minimal. Therefore, connector FEXT Loss requirements are used to model ACR-F performance for links and channels.

[d] For alternative measurement methods, see normative Annex H.

5.5.9 Power sum FEXT loss (for information only)

Conditions: Mated connectors, between all combinations of two pairs of contacts. See Table 19 below.

Table 19 – Power sum FEXT loss

Frequency MHz	Minimum PS FEXT loss [a] [b] dB		Test method
	Connector category		
	Category 7_A [c]	Category 8.2	
$1 \leq f \leq 1\ 000$	100,9 – 20 lg10(f)		IEC 60512-28-100 Test 28d
$1\ 000 \leq f \leq 2\ 000$		40,9 – 60 lg(f/1 000)	

[a] PS FEXT loss values that correspond to calculated values of greater than 72,0 dB shall revert to a minimum requirement of 72,0 dB.

[b] For connectors, the difference between PS FEXT loss and PS ACR-F is minimal. Therefore, connector PS FEXT loss requirements are used to model PS ACR-F performance for links and channels.

[c] For alternative measurement methods, see normative Annex H.

This characteristic is achieved by compliance to FEXT (5.5.8) and there is no necessity to test. The Power sum FEXT loss parameter shall be met by design.

5.5.10 Transverse conversion loss

Conditions: Mated connectors. See Table 20 below.

Table 20 – Transverse conversion loss

Frequency MHz	Minimum transverse conversion loss (TCL) [a] dB		Test standard
	Connector category		
	Category 7_A [b]	Category 8.2	
$1 \leq f \leq 1\ 000$	68 – 20 lg(f)		IEC 60512-28-100 test 28f
$1 \leq f \leq 2\ 000$		74 – 20 lg(f)	

[a] TCL values that correspond to calculated values of greater than 50,0 dB shall revert to a minimum requirement of 50,0 dB.

[b] For alternative measurement methods, see normative Annex I.

5.5.11 Transverse conversion transfer loss

Conditions: Mated connectors. See Table 21 below.

Table 21 – Transverse conversion transfer loss

Frequency MHz	Minimum transverse conversion transfer loss ($TCTL$) [a] dB		Test method
	Connector category		
	Category 7_A [b]	Category 8.2	
$1 \leq f \leq 1\ 000$	68 – 20 lg(f)		IEC 60512-28-100 test 28g
$1 \leq f \leq 2\ 000$		78 – 20 lg(f)	

[a] $TCTL$ values that correspond to calculated values of greater than 50,0 dB shall revert to a minimum requirement of 50,0 dB.

[b] For alternative measurement methods, see normative Annex I.

5.5.12 Power sum alien (exogenous) NEXT loss

Conditions: Mated connectors, between all combinations of two pairs of contacts. See Table 22 below.

Table 22 – Power sum alien (exogenous) NEXT loss

Frequency MHz	**Minimum power sum alien near end crosstalk loss** **(PS ANEXT)** dB **Connector category**		**Test method**
	Category 7_A	**Category 8.2**	
$1 \leq f \leq 1\ 000$	$125{,}5 - 20\ \lg(f)$ [a]		IEC 60512-25-9
$1 \leq f \leq 2\ 000$		$135{,}5 - 20\ \lg(f)$ [b]	

[a] PS ANEXT values that correspond to calculated values of greater than 72,0 dB shall revert to a minimum requirement of 72,0 dB.

[b] PS ANEXT values that correspond to calculated values of greater than 84,0 dB shall revert to a minimum requirement of 84,0 dB.

5.5.13 Power sum alien (exogenous) FEXT loss

Conditions: Mated connectors, between all combinations of two pairs of contacts. See Table 23 below.

Table 23 – Power sum alien (exogenous) FEXT loss

Frequency MHz	**Minimum power sum alien far end crosstalk loss** **(PS AFEXT)** [c] dB **Connector category**		**Test method**
	Category 7_A	**Category 8.2**	
$1 \leq f \leq 1\ 000$	$100{,}9 - 20\ \lg10(f)$ [a]		IEC 60512-25-9
$1 \leq f \leq 2\ 000$		$40{,}9 - 60\ \lg10(f/1\ 000)$ [b]	

[a] PS AFEXT values that correspond to calculated values of greater than 72,0 dB shall revert to a minimum requirement of 72,0 dB.

[b] PS AFEXT values that correspond to calculated values of greater than 84,0 dB shall revert to a minimum requirement of 84,0 dB.

[c] For connectors, the difference between PS AFEXT and PS AACR-F is minimal. Therefore, connector PS AFEXT requirements are used to model PS AACR-F performance for links and channels.

5.5.14 Coupling attenuation

Conditions: Mated connectors, all pairs of contacts. See Table 24 below.

Table 24 – Coupling attenuation

Frequency MHz	**Minimum coupling attenuation** dB		**Test method**
	Connector category		
	Category 7_A	**Category 8.2**	
$30 \le f \le 100$	85	85	IEC 62153-4-12
$100 < f \le$ Note	$85 - 20\ \lg(f)$	$85 - 20\ \lg(f)$	
NOTE Coupling attenuation is measured to 2 000 MHz but the limit applies to the upper frequency of the category under test.			

5.5.15 Transfer impedance

Conditions: Mated connectors, terminated with each cable construction intended to be used with these connectors. See Table 25 below.

Table 25 – Transfer impedance

Frequency MHz	**Maximum transfer impedance** Ω		**Test standard**
	Connector category		
	Category 7_A	**Category 8.2**	
$1 \le f \le 10$	$0{,}05\,f^{\,0{,}3}$	$0{,}05\,f^{\,0{,}3}$	IEC 60512-26-100 test 26e
$10 < f \le 80$	$0{,}01\,f$	$0{,}01\,f$	
$80 \le f \le 1\,000$		N/A[a]	
$1\,000 < f \le 2\,000$		N/A[a]	
[a] Transmission testing above 80 MHz does not apply (N/A).			

5.6 Mechanical characteristics

The following requirements shall apply to all variants:

a) mechanical operation
   – conditions: IEC 60512-9-1, Test 9a
   – speed: 10 mm/s maximum
   – rest: 1 s minimum (mated and unmated)
   – PL1: 750 operations
   – PL2: 2 500 operations
b) effectiveness of connector coupling devices
   – conditions: IEC 60512-15-6, Test 15f
   – mated connectors
   – all types: 50 N for 60 s minimum
c) insertion and withdrawal forces
   – conditions: IEC 60512-13-2, Test 13b

- speed: 10 mm/s maximum
- all types, insertion and withdrawal: 20 N maximum

6 Test schedule

6.1 General

The detail specification states the test sequence (in accordance with detail specification), and the number of specimens for each test sequence.

Individual variants shall be submitted to type tests for approval of those particular variants.

It shall be permissible to limit the number of variants tested to a selection representative of the whole range for which approval is required (which may be less than the range covered by the detail specification), but each feature and characteristic shall be proven.

Unless otherwise specified, mated sets of connectors shall be tested. For contact resistance measurements, care shall be taken to keep a particular combination of connectors together during the complete test sequence, that is, when unmating is necessary for a certain test, the same connectors shall be mated for subsequent tests.

6.2 Test procedures and measuring methods

The test methods specified and given in the relevant standards shall be the preferred methods but not necessarily the only ones that can be used. In case of dispute, however, the specified method shall be used as the reference method.

Unless otherwise specified, all tests shall be carried out under standard atmospheric conditions for testing as specified in IEC 60068-1.

Where approval procedures are involved and alternative methods are employed, it shall be the responsibility of the manufacturer to satisfy the authority granting approval that any alternative methods which he may use give results equivalent to those obtained by the methods specified.

6.3 Preconditioning

Before the tests are made, the connectors shall be preconditioned under standard atmospheric conditions for testing as specified in IEC 60068-1 for a period of 24 hours unless otherwise specified by the detail specification.

6.4 Wiring and mounting of specimens

6.4.1 Wiring

Wiring of these connectors shall take into account wire diameter of the cables defined in IEC 61156-2, IEC 61156-3, IEC 61156-4, IEC 61156-5, IEC 61156-6, IEC 61156-7, IEC 61156-9 and IEC 61156-10 as applicable. Where wiring and/or shielding of test specimens is required, the detail specification shall contain information suitable to conform to the selected methods of test.

6.4.2 Mounting

When mounting is required in a test, unless otherwise specified, the connectors shall be rigidly mounted on a metal plate or to specified accessories, whichever is applicable, using the specified connection methods, fixing devices and panel cut-outs as specified in 4.7.

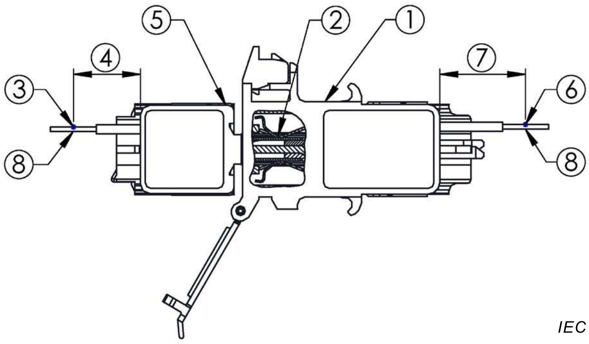
6.4.3 Arrangement for contact resistance measurement and procedure

6.4.3.1 General

The arrangement and procedure for contact resistance measurement shall be as given herein.

6.4.3.2 Arrangement for contact resistance measurement

For the measurement of contact resistance, the points of connection shall be as shown in Figure 25.



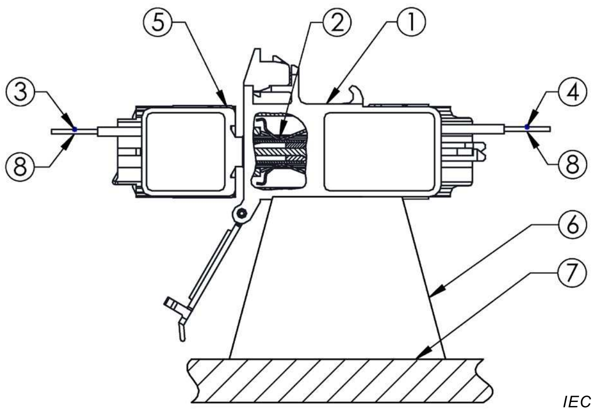
Key

1. Fixed connector.
2. Point B.
3. Point A.
4. As short as practical (except for vibration test CP1, see Table 32).
5. Free connector.
6. Point C.
7. As short as practical (except for vibration test CP1, see Table 32).
8. Contact resistance measurement points.

Figure 25 – Arrangement for contact resistance measurement

6.4.4 Arrangement for dynamic stress tests (test phase CP1)

Figure 26 shows arrangement for dynamic stress test.



Key

1. Fixed connector vibration feature.
2. Contact point.
3. Point A: secure to non-vibrating member.
4. Point C: secure to non-vibrating member.
5. Free connector.
6. Fixed connector rigidly fixed to the mounting plate.
7. Mounting plate.
8. Contact resistance measurement points.

Figure 26 – Arrangement for dynamic stress

6.5 Test schedules

6.5.1 General

The test parameters required shall not be less than those listed in 6.5.3.

6.5.2 Basic (minimum) test schedule

Not specified

6.5.3 Full test schedule

6.5.3.1 General

The following tests specify the characteristics to be checked and the requirements to be fulfilled.

For a complete test sequence, a minimum of 50 + N specimens are needed. This equals 5 groups of 10 for test group AP, BP, CP, DP, EP and FP. The N groups of 1 shall be for the “shield performance” testing, group GP. The N stands for each cable shield construction type the connectors are intended to be used for. Test group GP shall be performed N times with the specimens terminated with the different cable construction types the connectors are intended to be used for.

Where not otherwise stated, contact resistance test, including screen contacts, shall apply only to the interface between free connector and fixed connector (see 5.4).

6.5.3.2 Test group P preliminary

All test group specimens shall be subjected to the following tests given in Table 26. The specimens shall be tested in the following sequence as described within Table 26.

Table 26 – Test group P

Test phase	Test			Measurement to be performed		
	Title	**IEC 60512 Test No.**	**Severity or condition of test**	**Title**	**IEC 60512 Test No.**	**Requirements**
P 1	General examination	1		Visual examination	1a (IEC 60512-1-1)	There shall be no defects that would impair normal operation
				Examination of dimensions and mass	1b (IEC 60512-1-2)	The dimensions shall conform with those specified in the detail specification
P 2	Polarization					
P 3	Contact resistance		Measurement points as in Figure 25 All contacts/specimens	Millivolt level method or contact resistance – specified test current method	2a (IEC 60512-2-1)	20 mΩ maximum change from initial for signal contacts. 100 mΩ maximum input to output resistance for shield
P 4			100 V ± 15 V d. c.	Insulation resistance	3a (IEC 60512-3-1)	500 MΩ minimum
P 5			One signal contact to all other signal contacts connected together. Method A Mated connectors	Voltage proof	4a (IEC 60512-4-1)	1 000 V DC or AC peak
			All signal contacts connected together to shield,(housing/ mounting plate) if present Method A Mated connectors			1 500 V DC or AC peak

6.5.3.3 Test group AP

See Table 27 for test phases AP1 through AP15.

Table 27 – Test group AP

Test phase	Test			Measurement to be performed		
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	Requirements
AP 1	Insertion and withdrawal forces	13b (IEC 60512-13-2)	Connector locking device depressed			Insertion force 20 N max Withdrawal force 20 N max
AP 2	Effectiveness of connector coupling device	15f (IEC 60512-15-6)	Rate of load application 44,5 N/s maximum			50 N for 60 s ± 5 s
AP 3	Rapid change of temperature	11d (IEC 60512-11-4)	–40°C to 70°C Mated connectors 25 cycles ι = 30 min Recovery time 2 h			
AP 4			Test voltage 100 V ± 15 V direct current Method A Mated connectors	Insulation resistance	3a (IEC 60512 -3-1)	500 MΩ minimum
AP 5			Measurement points as in Figure 25 All contacts/specimens	Contact resistance	2a (IEC 60512 -2-1)	20 mΩ maximum change from initial for signal contacts. 100 mΩ maximum input to output resistance for shield
AP 6			One signal contact to all other signal contacts connected together. Method A Mated connectors	Voltage proof	4a (IEC 60512 -4-1)	1 000 V DC or AC peak
			All signal contacts connected together to shield,(housing/ mounting plate) if present Method A Mated connectors			1 500 V DC or AC peak
AP 7			Unmated connectors	Visual examination	1a (IEC 60512 -1-1)	There shall be no defects that would impair normal operation.
AP 8	Damp heat, cyclic	IEC 60068-2-38. test Z/AD	21 cycles low temperature 25 °C high temperature 65 °C cold sub-cycle – 10 °C humidity 93 % Half of the samples in mated state Half of the samples in unmated state			

Test phase	Test			Measurement to be performed		
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	Requirements
AP 9			Measurement points as in Figure 25 All Contacts/Specimens	Contact resistance	2a (IEC 60512 -2-1)	20 mΩ maximum change from initial for signal contacts. 100 mΩ maximum input to output resistance for shield.
AP 10	Insertion and withdrawal forces	13b (IEC 60512-13-2)	Connector locking device depressed			Insertion force 20 N max Withdrawal force 20 N max
AP 11	Effectiveness of connector coupling device	15f (IEC 60512-15-6)	Rate of load application 44,5 N/s maximum			50 N for 60 s $\pm$ 5 s
AP 12			Unmated connectors	Visual examination	1a (IEC 60512 -1-1)	There shall be no defects that would impair normal operation.
AP 13	Solderability		As applicable			
AP 14	Resistance to soldering heat		As applicable			
AP 15			One signal contact to all other signal contacts connected together. Method A Mated connectors	Voltage proof	4a (IEC 60512 -4-1)	1 000 V DC or AC peak
			All signal contacts connected together to shield,(housing /mounting plate) if present Method A Mated connectors			1 500 V DC or AC peak
Do not perform step AP 15 if solderability and resistance to soldering heat were not performed.						

6.5.3.4 Test group BP

See Table 28 for test phases BP1 through BP9.

Table 28 – Test group BP

Test phase	Test			Measurement to be performed		
	Title	**IEC 60512 Test No.**	**Severity or condition of test**	**Title**	**IEC 60512 Test No.**	**Requirements**
BP 1	Locking device mechanical operations		2 N operations – see mechanical operations			See Annex B
BP 2	Mechanical operations	9a (IEC 60512-9-1)	N/2 operations see mechanical operations. Speed 10 mm/s Rest 1 s (when mated and unmated) Locking device inoperative			N is: PL1 = 750 PL2 = 2 500
BP 3	Flowing mixed gas corrosion	11g (IEC 60512-11-7)	Method 1 4 days Half of the samples in mated state Half of the samples in unmated state		11g (IEC 60512-11-7)	
BP 4			Measurement points as in Figure 25 All contacts/specimen	Contact resistance	2a (IEC 60512-2-1)	20 mΩ maximum change from initial for signal contacts. 100 mΩ maximum input to output resistance for shield
BP 5	Mechanical operations	9a (IEC 60512-9-1)	N/2 operations see mechanical operations. Speed 10 mm/s Rest 1 s (when mated and unmated) Locking device inoperative			
BP 6			Measurement points as in Figure 25 All contacts/specimen	Contact resistance	2a (IEC 60512-2-1)	20 mΩ maximum change from initial
BP 7			100 V ± 15 V direct current Method A Mated connectors	Insulation resistance	3a (IEC 60512-3-1)	500 MΩ minimum
BP 8			One signal contact to all other signal contacts connected together. Method A Mated connectors	Voltage proof	4a (IEC 60512-4-1)	1 000 V DC or AC peak
			All signal contacts connected together to shield,(housing/ mounting plate) if present Method A Mated connectors			1 500 V DC or AC peak

Test phase	Test			Measurement to be performed		
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	Requirements
BP 9				Visual examination	1a (IEC 6051 2-1-1)	There shall be no defects that would impair normal operation.

6.5.3.5 Test group CP

See Table 29 for test phases CP1 through CP4.

Table 29 – Test group CP

Test phase	Test			Measurement to be performed		
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	Requirements
CP 1	Vibration	6d (IEC 60512 -6-4)	F = 10 –500 Hz Amplitude = 0,35 mm Acceleration = 50 m/s 10 sweeps per/ axis Measure points as in Figure 26	Contact disturbance	2e (IEC 60512-2-5)	Interruption of <10 µs
CP 3			Measurement points as in Figure 25 All contacts/specimens	Contact resistance	2a (IEC 6051 2-2-1)	20 mΩ maximum change from initial for signal contacts. 100 mΩ maximum input to output resistance for shield
CP 2			Test voltage 100 V ± 15 V direct current Method A Mated connectors	Insulation resistance	3a (IEC 6051 2-3-1)	500 MΩ Minimum
CP 4			Unmated connectors	Visual examination	1a (IEC 6051 2-1-1)	There shall be no defects that would impair normal operation

6.5.3.6 Test group DP

See Table 30 for test phases DP1 through DP7.

Table 30 – Test group DP

Test phase	Test			Measurement to be performed		
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	Requirements
DP 1	Electrical load and temperature	9b (IEC 60512 -9-2)	5 connectors 500 h 70°C Recovery period 2 h			0,5 A 5 connectors no current 5 connectors
DP 2			Test voltage 100 V ± 15 V direct current Method A Mated connectors	Insulation resistance	3a (IEC 6051 2-3-1)	500 MΩ Min.
DP 3			One signal contact to all other signal contacts connected together. Method A Mated connectors	Voltage proof	4a (IEC 6051 2-4-1)	1 000 V DC or AC peak
			All signal contacts connected together to shield, (housing/ mounting plate) if present Method A Mated connectors			1 500 V DC or AC peak
DP 4			Unmated connectors	Visual examination	1a (IEC 6051 2-1-1)	There shall be no defects that would impair normal operation
DP 5			Measurement points as in Figure 25 All contacts/specimens	Contact resistance	2a (IEC 6051 2-2-1)	20 mΩ maximum change from initial for signal contacts. 100 mΩ maximum input to output resistance for shield
DP6	Mechanical gauging		Both, free and fixed connector			Passing Go/No go test
DP7	Gauging continuity	Annex A of this document	All signal contacts and shield / specimens	Contact disturbance	2e (IEC 6051 2-2-5)	Interruption of <10 µs

6.5.3.7 Test group EP

See Table 31 for test phases EP1 through EP11.

Table 31 – Test group EP

Test phase	Test			Measurement to be performed		
	Title	IEC 60512 Test No.	Severity or condition of test	Title	IEC 60512 Test No.	Requirements
EP1				Insertion loss	Annex E of this document or 60512-28-100 test 28a	See 5.5.2
EP 2			All pairs, both directions (pair to pair)	NEXT loss	Annex G of this document or 60512-28-100 test 28c	See 5.5.6
EP 3			All pairs, both directions	Return loss	Annex F of this document or 60512-28-100 test 28b	See 5.5.3
EP 4			All pairs, both directions (pair to pair)	FEXT loss	Annex H of this document or 60512-28-100 test 28d	See 5.5.8
EP 5				Transverse conversion loss	Annex I of this document or 60512-28-100 test 28f	See 5.5.10
EP6				Transverse conversion transfer loss	Annex I of this document or 60512-28-100 test 28g	See 5.5.11
EP7				Transfer impedance	IEC 60512-26-100	See 5.5.15
EP8			All pairs, both directions, (pair to pair)	ANEXT loss	Annex G of this document or 60512-25-9	See 5.5.12
EP9			All pairs, both directions, (pair to pair)	AFEXT loss	Annex H of this document or 60512-25-9	See 5.5.13
EP 10	Input to output resistance		Measurement points as defined in 5.4.5 All signal contacts and switch contacts / specimens	Millivolt level method	2a (IEC 60512-2-1)	Signal contact resistance= 200 mΩ maximum Shield resistance = 100 mΩ maximum
EP 11	Input to output resistance unbalance		Measurement points as defined in 5.4.6 All signal contacts and switch contacts / specimens	Millivolt level method	2a (IEC 60512-2-1)	Unbalance resistance = 50 mΩ maximum
All measurements shall be performed on mated connectors.						

6.5.3.8 Test group FP

See Table 32 for test phases FP1 through FP3.

Table 32 – Test group FP

Test phas e	Test			Measurement to be performed		
	Title	**IEC 60512 Test No.**	**Severity or condition of test**	**Title**	**IEC 60512 Test No.**	**Requirements**
FP 1	Surge test	ITU-T K.20	Contact/contact unexposed environments Mated connectors tests 1, 2 and 3			Test 2.1 and 2.2: Acceptance criteria A per ITU-T K.44:2012,Clause 9, Test 2.3: Acceptance criteria per ITU-T K.44:2012,Clause 9,
FP 2			100V $\pm$ 15 V direct current Method A Mated connectors	Insulation resistance	3a (IEC 60512-3-1)	500 MΩ min
FP 3			Unmated connectors	Visual examination	1a (IEC 60512-1-1)	There shall be no defects that would impair normal operation

6.5.3.9 Test group GP

See Table 33 for test phases GP1 through GP5.

Table 33 – Test group GP

Test phas e	Test			Measurement to be performed		
	Title	**IEC 60512 Test No.**	**Severity or condition of test**	**Title**	**IEC 60512 Test No.**	**Requirements**
GP 1	High Temperature	9b (IEC 6051 2-9-2)	500 h 70 °C Recovery period 2 h			
GP 2	Cyclic damp heat	IEC 60068 -2-38. test Z/AD	21 cycles low temperature 25 °C high temperature 65 °C cold sub-cycle – 10°C, humidity 93 % Half of the samples in mated state Half of the samples in unmated state			
GP 3			Additional test are for further study			
GP4				Transfer impedance	26-100 test 26e	See 5.5.15
GP5				Coupling attenuation	IEC 62153-4-12	See 5.5.14

Annex A
(normative)

Gauging procedure

A.1 Fixed connectors

The "Go" gauge shall be capable of being inserted and removed with a force of 8,9 N maximum.

The "No-Go" gauges shall not be capable of entering the fixed connector more than 1,78 mm with an 8,9 N insertion force.

A.2 Free connectors

The connector shall be capable of insertion and latching into the "Go" gauge with a 20 N or less insertion force with the latch bar depressed.

After insertion and latching, the connector shall be capable of removal, with the latch depressed, with a removal force of 20 N or less applied at an advantageous angle.

The free connectors shall not be capable of entering the "No-Go" gauges more than 1,78 mm with an 8,9 N insertion force.

Annex B
(normative)

Locking device mechanical operation – test procedure and requirements

B.1 Object

The object of this mechanical endurance test shall be to assess the operational limits of the locking device on the free connectors.

B.2 Preparation of the specimens

The specimen shall be prepared and mounted so that the locking device is readily accessible for application of the test. No other movement of the free connector shall be allowed.

B.3 Test method

The specimen shall be subjected to mechanical operational endurance tests of the number of cycles, as specified in Test group BP, BP1, Table 28.

The speed of the operation of the applied force to the locking device shall not exceed 20 cycles per minute.

The specimen shall be operated in the normal manner, and the locking device shall be depressed until it contacts the body of the free connector.

Mechanical aids, which simulate normal operations, may be used, provided that they do not introduce abnormal stresses.

B.4 Final measurements

After the specified number of operations, the specimens shall show no visual indication of fatigue or stress cracking of the locking device when examined per IEC 60512-1-1 test 1a.

Annex C
(normative)

Plug and outlet interoperability qualification

C.1 Object

Annex C defines the test procedure for ensuring the IEC 61076-3-104 plug and outlet used to quantify mated performance meet the minimum transmission requirements separately.

C.2 Test equipment

The equipment used shall be as described in Annex D.

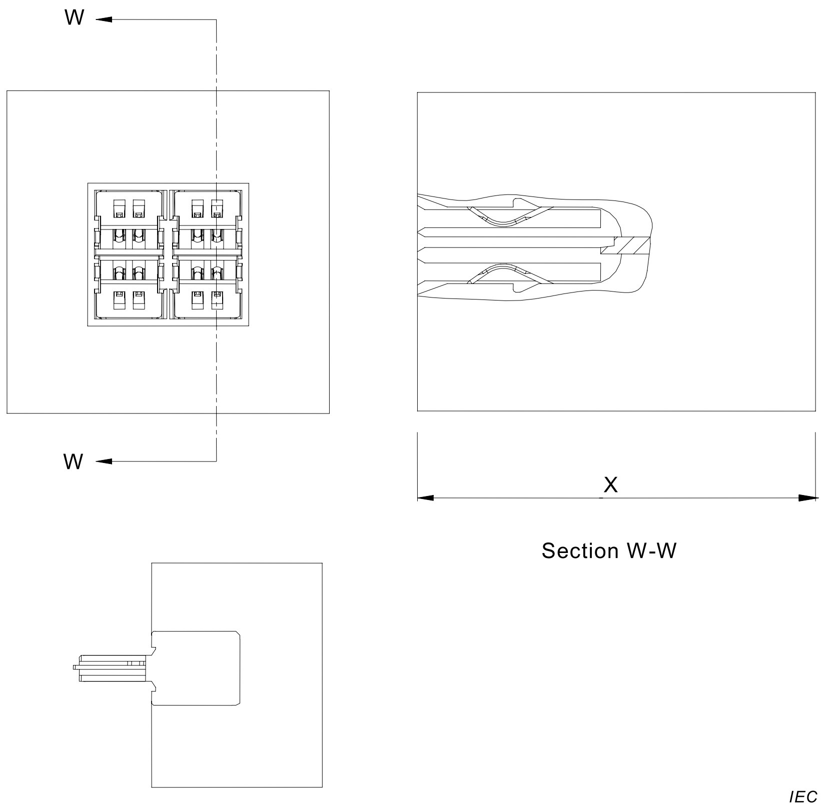


Figure C.1 – Precision test fixtures (covers)

The precision free connector test fixture “cover” and fixed connector test fixture “cover” shall be used to shield the terminations on the far side of the connector under test.

The test covers shall be similar to those shown above in Figure C.1. The internal dimensions are as shown in 4.2.2 and 4.2.4 of this document. The fixture consists of an outlet with a cover over the termination area that completely isolates the pairs. The cover portion (dimension X) may vary from manufacturer to manufacturer due to overall length of connector.

An alternate fixture using the PCB outlet mounted on a printed circuit board with appropriate connections for connecting to a network analyser can also be used. A similar plug version on a printed circuit board can also be used.

C.3 Test procedure

With the connector mated to the appropriate test fixture, measure NEXT and FEXT performance for each pair combination in accordance with IEC 60512-28-100.

The performance of the precision test fixture, plug and outlet, are verified to have return loss, NEXT and FEXT performance, at minimum 6 dB superior to that of the standard performance requirements in test group EP.

Annex D
(normative)

General requirements for the measurement set-up

D.1 Test instrumentation

These electrical test procedures require the use of a vector network analyzer. The analyzer should have the capability of full two port calibrations. The analyser shall cover the frequency range of 1 MHz to 1 GHz at least.

At least 2 test baluns are required in order to perform measurements with balanced symmetrical signals. The requirements for the baluns are given in Clause D.4.

Reference loads and cables are needed for the calibration of the set-up. Requirements for the reference components are given in D.5.1 and D.5.2 respectively.

Termination loads are needed for termination of pairs, used and unused, which are not terminated by the test baluns. Requirements for the termination loads are given in D.6.

A test adapter (triaxial test set) is needed for the transfer impedance measurements. Reference to requirements for this set-up is given in IEC 60512-26-100.

An absorbing clamp and ferrite absorbers are needed for the coupling attenuation measurements. The requirements for these items are given in EN 50289-1-14, coupling attenuation test method.

D.2 Coaxial cables and test leads for network analysers

Coaxial cable assemblies between network analyser and baluns should be as short as possible. It is recommended that they do not exceed 60 cm each.

The baluns shall be electrically bonded to a common ground plane. For crosstalk measurements, a test fixture may be used, in order to reduce residual crosstalk (see Annex J).

Balanced test leads and associated connecting hardware to connect between the test equipment and the connector under test shall be taken from components that meet or exceed the requirements for the relevant category. Balanced test leads shall be limited to maximum 7 cm between each balun and the reference plane of the connector under test. Pairs shall remain twisted from the baluns to where connections are made.

D.3 Measurement precautions

To assure a high degree of reliability for transmission measurements, the following precautions are required:

a) Consistent and stable balun and resistor loads shall be used for each pair throughout the test sequence.
b) Cable and adapter discontinuities, as introduced by physical flexing, sharp bends and restraints shall be avoided before, during and after the tests.
c) The relative spacing of conductors in the pairs shall be preserved throughout the tests to the greatest extent possible.

d) The balance of the cables is maintained to the greatest extent possible by consistent conductor lengths and pair twisting to the point of load.

e) The sensitivity to set-up variations for these measurements at high frequencies demands attention to details for both the measurement equipment and the procedures.

D.4 Balun requirements

The baluns may be balun transformers or 180° hybrids with attenuators to improve matching if needed (see Figure D.1).

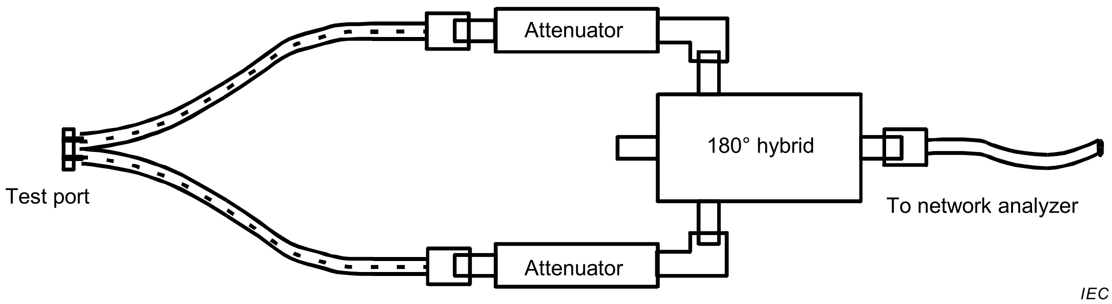


Figure D.1 – 180° hybrid used as a balun

The specifications for the baluns shall apply for the whole frequency range for which they are used. Baluns shall be RFI shielded and shall conform to the specifications listed in Table D.1.

Table D.1 – Test balun performance characteristics

Parameter	**Requirement at test frequencies up to** 250 MHz	**Requirement at test frequencies above** 250 MHz
Impedance, primary	Matched to applied network analyser	
Impedance, secondary	100 Ω	
Insertion loss	10 dB maximum	
Return loss secondary	14 dB minimum	
Return loss common mode with common mode termination [a]	10 dB minimum	
Return loss common mode without common mode termination [a]	1 dB maximum	NA
Longitudinal balance [b]	50 dB	NA
Common mode rejection [c]	50 dB	40 dB
Output signal balance [c]	50 dB	40 dB
Power rating	0,1 W	

[a] NA: not applicable

[b] Measured by connecting the balanced output terminals together and measuring the return loss. The nominal primary impedance shall terminate the primary input terminal. Applicable for baluns, which are used for balance measurements. Measured from primary input terminal to common mode terminal when secondary balanced terminal is terminated with 100 Ω.

[C] Measured according to ITU-T Recommendations G.117 and O.9.

D.5 Reference components for calibration

D.5.1 Reference loads for calibration

To perform a one or two-port calibration of the test equipment, a short circuit, an open circuit and a reference load are required. These devices shall be used to obtain a calibration at the reference plane.

The reference load shall be calibrated against a calibration reference, which shall be a 50 Ω load, traceable to an international reference standard. Two 100 Ω reference loads in parallel shall be calibrated against the calibration reference. The reference loads for calibration shall be placed in an SMA type connector according to IEC 60169-15 meant for panel mounting, which is machined flat on the back side (see Figure D.2). The loads shall be fixed to the flat side of the connector, distributed evenly around the centre conductor. A network analyser shall be calibrated, one port full calibration, with the calibration reference. Thereafter the return loss of the reference loads for calibration shall be measured. The verified return loss shall be greater than 46 dB at frequencies up to 100 MHz and greater than 40 dB at frequencies above 100 MHz and up to the limit for which the measurements are to be carried out.

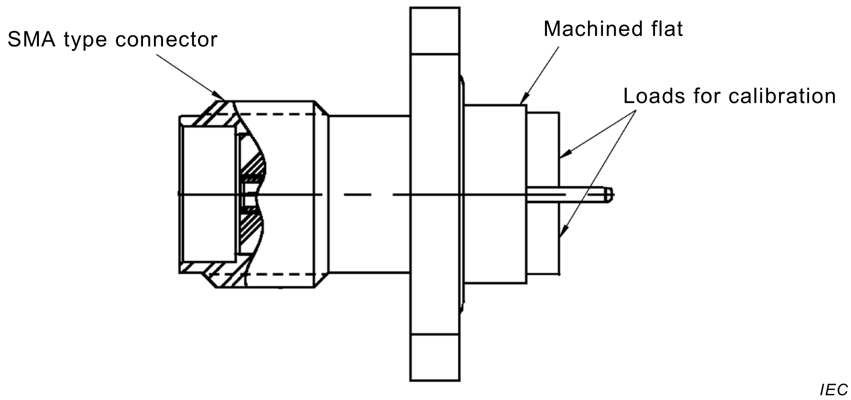


Figure D.2 – Calibration of reference loads

D.5.2 Reference cables for calibration

As a minimum, reference cable that is used to perform calibration of the test set-up shall satisfy the requirement of the same category according to IEC 61156 (all parts) as the category of the connector. The reference cable shall be a length of horizontal cable for which the sheath is preserved. One of the pairs of the reference cable is used for the calibrations. The total length of reference cable shall be according to the length of the measurement cables as outlined in the calibration procedures for the various tests. Both ends of the reference cable shall be well prepared, so that the twisting is maintained up to the test ports.

D.6 Termination loads for termination of conductor pairs

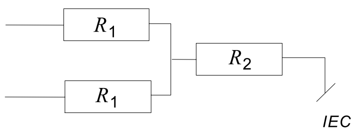
During measurement, conductor pairs of the measurement cables for the connector under test shall be terminated according to the specified test set-up with impedance matching loads. For pairs under test this is provided by the test instrumentation at one or both ends. For pairs not under test or not connected to test instrumentation, resistor loads or terminated baluns shall be applied. For differential mode only terminations, only resistor loads are allowed[2].

2 Unpredictable stray capacitances in baluns cause resonances at high frequencies if they are used as terminations when the common mode terminal is open.

The nominal differential mode impedance of the termination shall be 100 Ω. The nominal common mode impedance shall be 50 Ω ± 25 Ω.

NOTE The exact value of the common mode impedance is not critical for most measurements. Normally, a value of 75 Ω is used for unscreened connectors while a value of 25 Ω is used for screened connectors.

Resistor loads shall use resistors specified for ±1 % accuracy at direct current and have a return loss greater than 40 –10lg (f) where f is the frequency in megahertz[3]. For pairs connected to a balun, common mode load is implemented by applying a load at the common mode terminal (centre tap) of the balun. The impedance of the load is equal to the common mode impedance. For a balun without a common mode terminal (centre tap is not accessible), the requirement for common mode return loss shall be complied with by inserting a balanced attenuator between the balun and the connector pair. Guidance on how this is done is shown in Annex J. For pairs connected to resistor loads, common mode load is implemented by the Y configuration shown in Figure D.3.



Key

$$R_1 = \frac{R_{dif}}{2}$$

and

$$R_2 = R_{com} - \frac{R_{dif}}{4}$$

where

R_{dif} is the differential mode impedance (Ω);

R_{com} is the common mode impedance (Ω).

Figure D.3 – Resistor load

The two resistors R_1 shall be matched to within 0,5 %. The termination shall be implemented at a small printed circuit board with surface mount resistors. The layout for the resistors R_1 shall be symmetrical.

The common mode termination points for all pairs shall be connected to the ground plane.

D.7 Termination of screens

If the connector under test is screened, screened measurement cables shall be applied.

The screen or screens of these cables shall be fixed to the ground plane as close as possible to the measurement baluns.

3 Return loss of terminations are measured with a network analyser connected to one balun, which is calibrated (full one port calibration) using the reference loads (see D.5.1).

D.8 Test specimen and reference planes

The test specimen is a mated pair of relevant connectors. The electrical reference plane for the test specimen is the point at which the cable sheath enters the connector (the back end of the connector), or the point, at which the internal geometry of the cable is no longer maintained, whichever is farther from the connector (see Figure D.4). This definition applies to both ends of the test specimen.

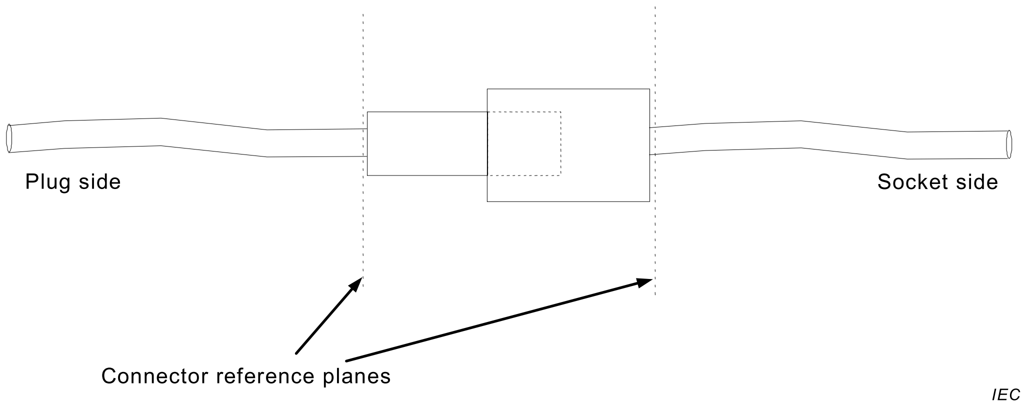


Figure D.4 – Definition of reference planes

Annex E
(normative)

Insertion loss

E.1 Object

The object of this test shall be to measure the insertion loss.

E.2 Test method

Insertion loss is evaluated by measuring the scattering parameters, S_{21}, of all the conductor pairs.

E.3 Test set up

The test set-up consists of a network analyser and two baluns as defined in Annex D.

It is not necessary to terminate the unused pairs.

E.4 Procedure

E.4.1 Calibration

A full 2-port calibration shall be performed at the reference plane. This is performed by applying a maximum length of 14 cm reference cable between the terminals of the baluns and performing the transmission calibration measurement. Then maximum lengths of 7 cm reference cables are connected to the terminals of the two baluns (see Figure E.1). The total length of these cables shall be equal to the length of the reference cable used for transmission calibrations. At the end of these reference cables, the reflection calibrations are performed by applying open, short and load terminations.

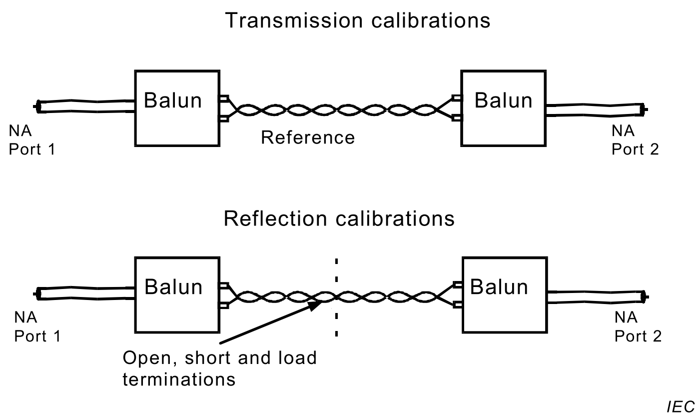


Figure E.1 – Calibration

E.4.2 Measurement

The test specimen shall be terminated with measurement cables at both ends (see Figure E.2). The length of measurement cables shall be equal to the length of the reference cables used for reflection calibrations. The measurement cables shall be the cable types for which the connector is intended. A S_{21} measurement shall be performed.

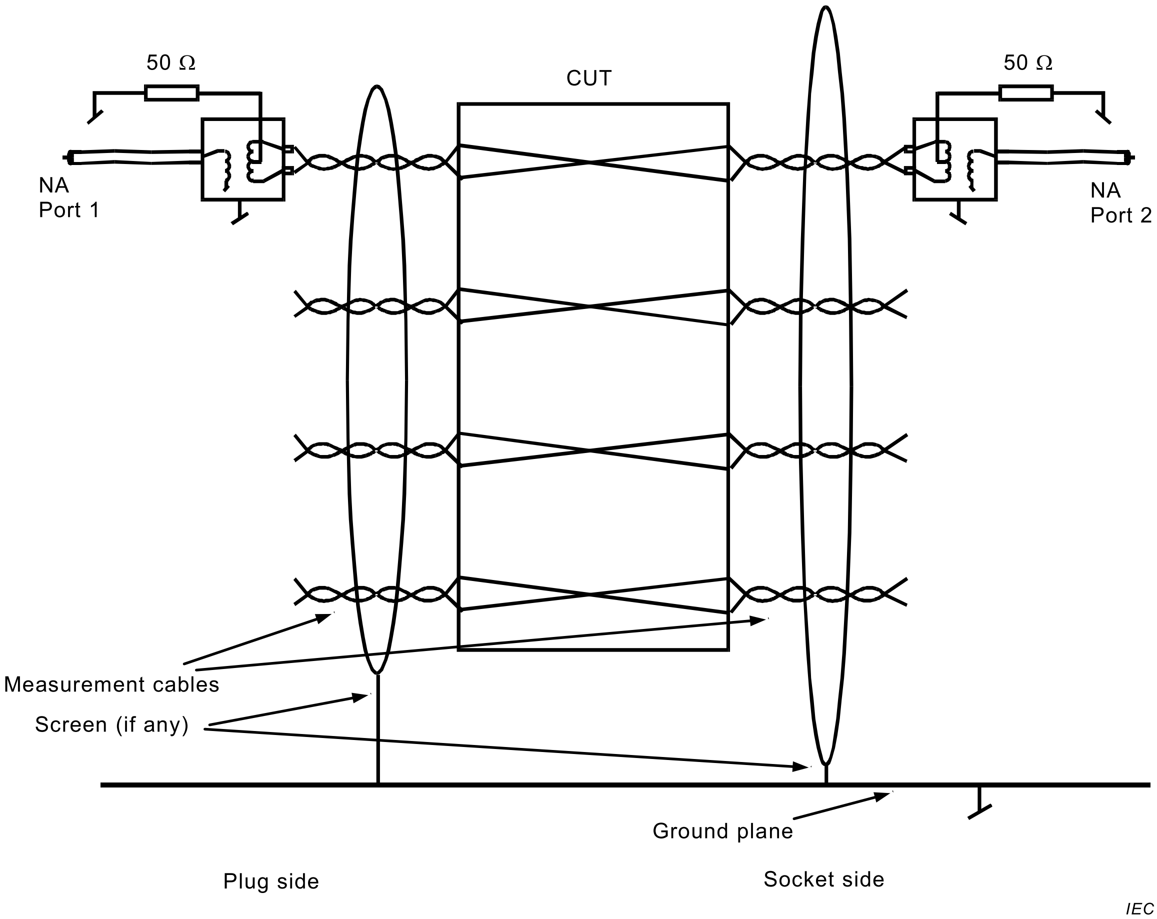


Figure E.2 – Measuring set-up

E.5 Test report

The measured results shall be reported in graphical or table format with the specification limits shown on the graphs or in the table at the same frequencies as specified in the relevant detail specification. Results for all pairs shall be reported. It shall be explicitly noted if the measured results exceed the test limits.

E.6 Accuracy

The insertion loss accuracy shall be within ±0,05 dB.

Annex F
(normative)

Return loss

F.1 Object

The object of this test shall be to measure the return loss of a mated connector pair at the two reference planes.

F.2 Test method

Return loss shall be measured by measuring the scattering parameters, S_{11} and S_{22} of all the conductor pairs.

NOTE As a connector is a low loss device, the return loss with respect to the two sides is nearly equal.

F.3 Test set-up

The test set-up shall be as described in Annex D.

F.4 Procedure

F.4.1 Calibration

Calibration shall be performed as described in E.4.1.

F.4.2 Measurement

The test specimen shall be terminated with measurement cables at both ends. The length of measurement cables shall be equal to the length of the reference cables used for reflection calibrations. The measurement cables shall be the cable types for which the connector is intended. S_{11} and S_{22} measurements shall be carried out for each of the pairs.

F.5 Test report

The measured results shall be reported in graphical or table format with the specification limits shown on the graphs or in the table at the same frequencies as specified in the relevant detail specification. Results for all pairs shall be reported. It shall be explicitly noted if the measured results exceed the test limits.

F.6 Accuracy

The return loss of the load for calibration is verified to be greater than 46 dB up to 100 MHz and greater than 40 dB at higher frequencies. The uncertainty of the connection between the connector under test and the baluns are expected to deteriorate the return loss of the set-up (effectively the directional bridge implemented by the test set-up) by 6 dB. The accuracy of the return loss measurements is then equivalent with measurements performed by a directional bridge with a directivity of 40 dB and 34 dB. The accuracy (uncertainty band) is tabled in Table F.1.

Table F.1 – Uncertainty band of return loss measurement at frequencies below 100 MHz

Measured RL (dB)	10	12	15	18	20	22	25	28	30
Lower uncertainty limit (dB)	–0,3	–0,3	–0,5	–0,7	–0,8	–1,0	–1,4	–1,9	–2,4
Higher uncertainty limit (dB)	+0,3	+0,4	+0,5	+0,7	+0,9	+1,2	+1,7	+2,5	+3,3

Annex G
(normative)

Near end cross talk (NEXT)

G.1 Object

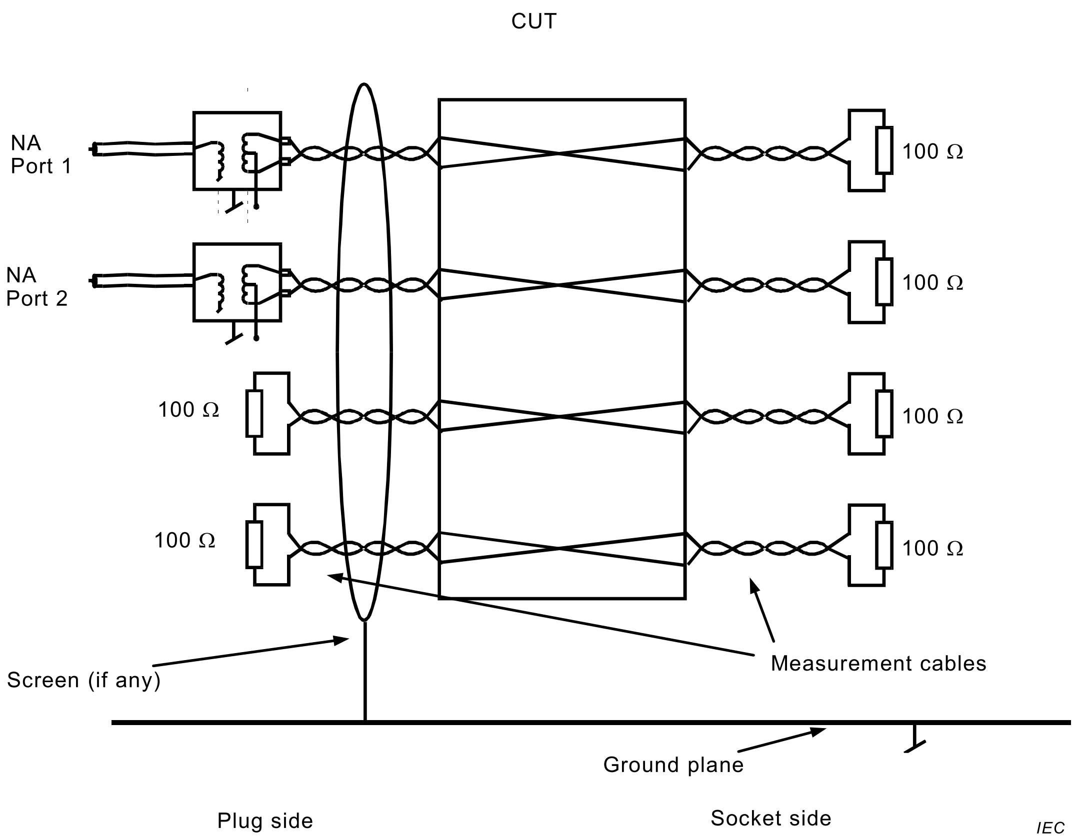
The object of this test procedure shall be to measure the magnitude of the electric and magnetic coupling between driven (disturbing) and quiet (disturbed) pairs of a mated connector pair.

G.2 Test method

Near end crosstalk shall be evaluated by measuring the scattering parameters, S_{21}, of the possible conductor pair combinations at one end of the mated connector, while the other end of the pairs is terminated.

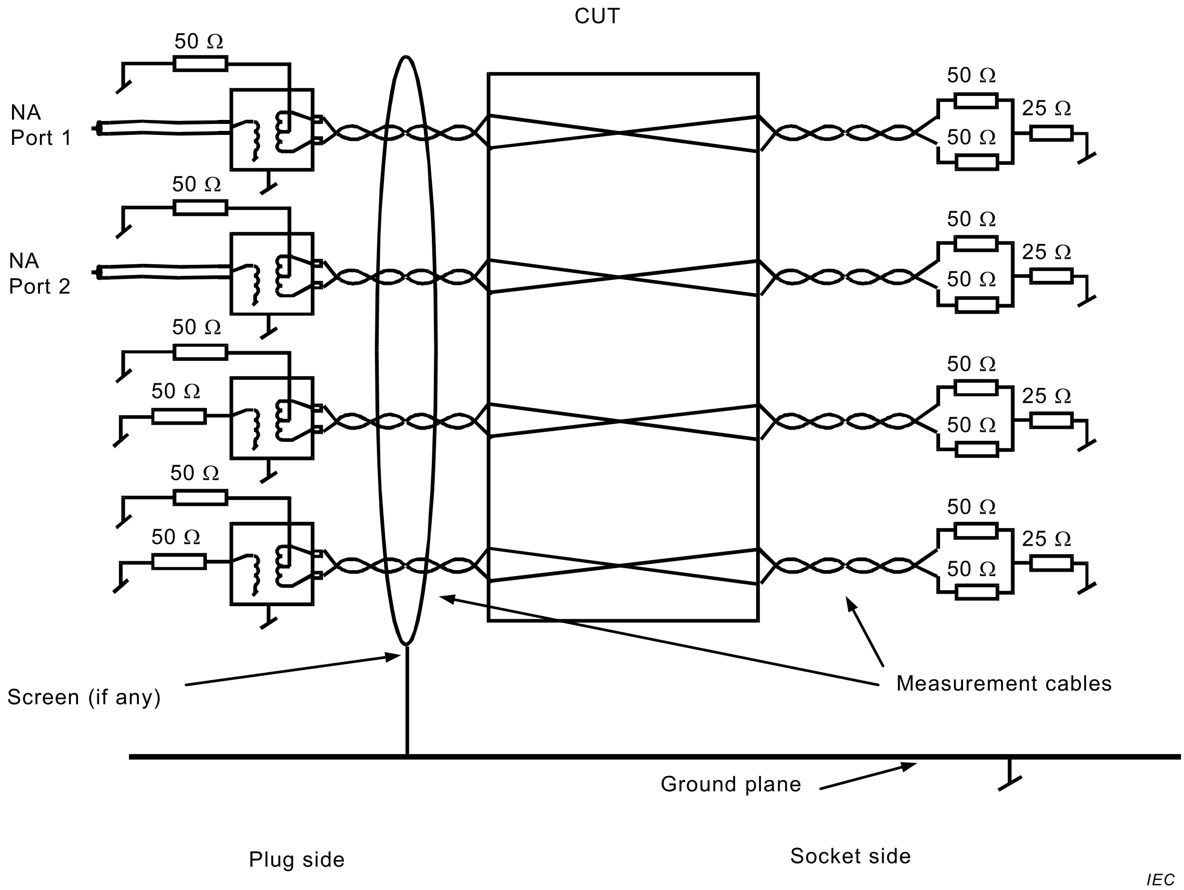
G.3 Test set-up

The test set-up consists of two baluns and a network analyser, shall be as given in Annex D. A representation of the set-up, which also shows the termination principles, is shown in Figures G.1.and G.2.



Passive terminations shall be resistor terminations.

Figure G.1 – NEXT measurement differential mode only terminations



Passive terminations may be either balun or resistor terminations.

Figure G.2 – NEXT measurement differential and common mode terminations

G.4 Procedure

G.4.1 Calibration

A through calibration shall be applied as a minimum. Full two port calibrations are recommended in order to enhance the measurement accuracy.

G.4.2 Establishment of noise floor

The noise floor of the set up shall be measured. The level of the noise floor is determined by white noise, which may be reduced by increasing the test power and by reducing the bandwidth of the network analyser, and by residual crosstalk between the test baluns. The noise floor shall be measured by terminating the baluns with resistors and perform a S_{21} measurement. The noise floor shall be 20 dB lower than any specified limit for the crosstalk. If the measured value is closer to the noise floor than 10 dB, this shall be reported.

If the test results have high crosstalk values, it may be necessary to screen the terminating resistors.

G.4.3 Measurement

The disturbing pair of the connector under test (CUT) shall be connected to the signal source and the disturbed pair to the receiver port. Termination shall be made according to Figure G.1 and Figure G.2. All possible pair combinations shall be tested and the results recorded.

It is recommended that the socket be terminated with short separated pairs without a cable jacket.

The CUT shall be tested in the following configurations:

a) With differential and common mode terminations only. For category 7 connectors this test is not required. This requirement is under consideration for all categories should experience show that a test for balance can replace this requirement.
b) The measurements shall be performed at both ends of the mated connector.

NOTE As a connector is a low loss device, near end cross talk values from the two ends are nearly equal.

G.5 Test report

The measured results shall be reported in graphical or table format with the specification limits shown on the graphs or in the table at the same frequencies as specified in the relevant detail specification. Results for all pairs shall be reported. It shall be explicitly noted if the measured results exceed the test limits.

G.6 Accuracy

The accuracy shall be better than ±1 dB at measurements up to 60 dB and ±2 dB at measurements up to 85 dB.

Annex H
(normative)

Far end cross talk (FEXT)

H.1 Object

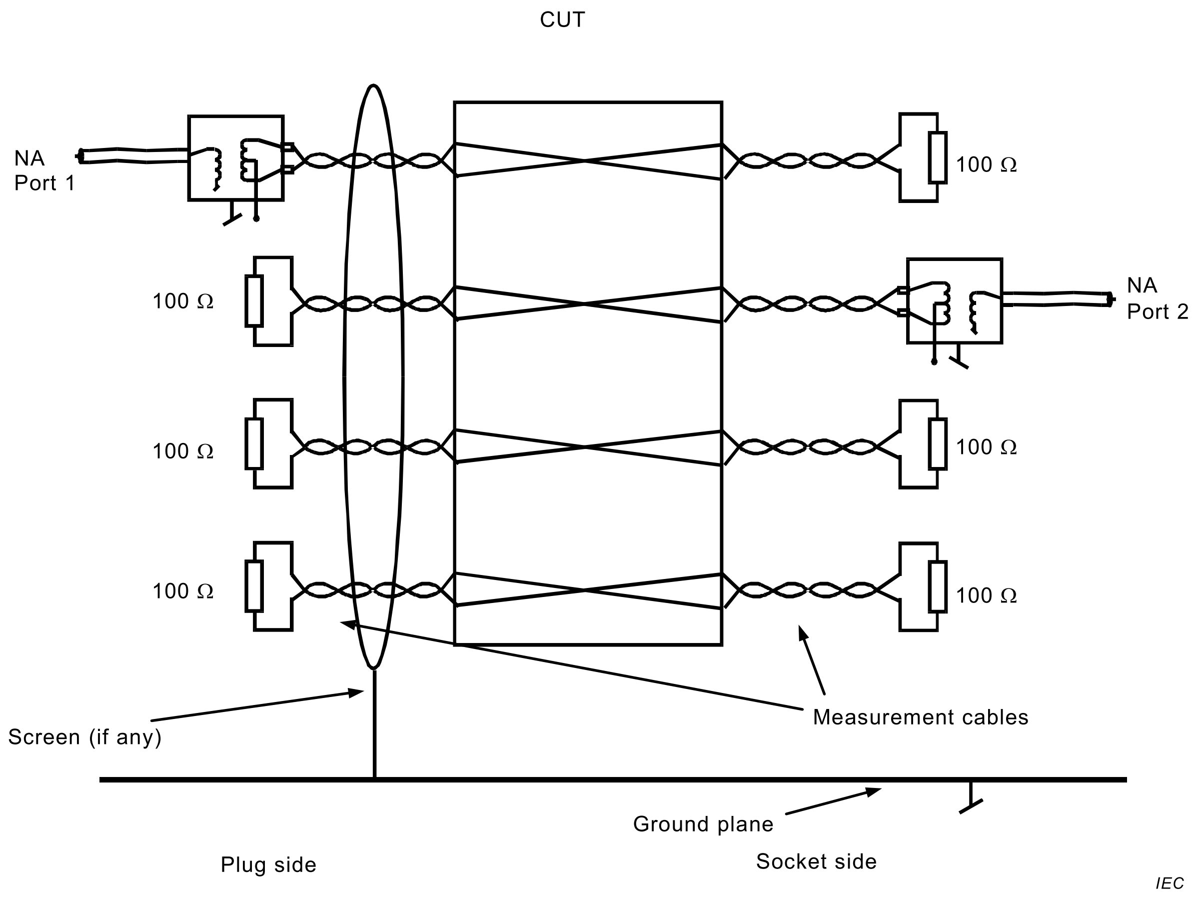
The object of this test procedure shall be to measure the magnitude of the electric and magnetic coupling between driven (disturbing) and quiet (disturbed) pairs of a mated connector pair.

H.2 Test method

Far end crosstalk shall be evaluated by measuring the scattering parameters, S_{21}, of the possible conductor pair combinations at one end of the mated connector, to the other end.

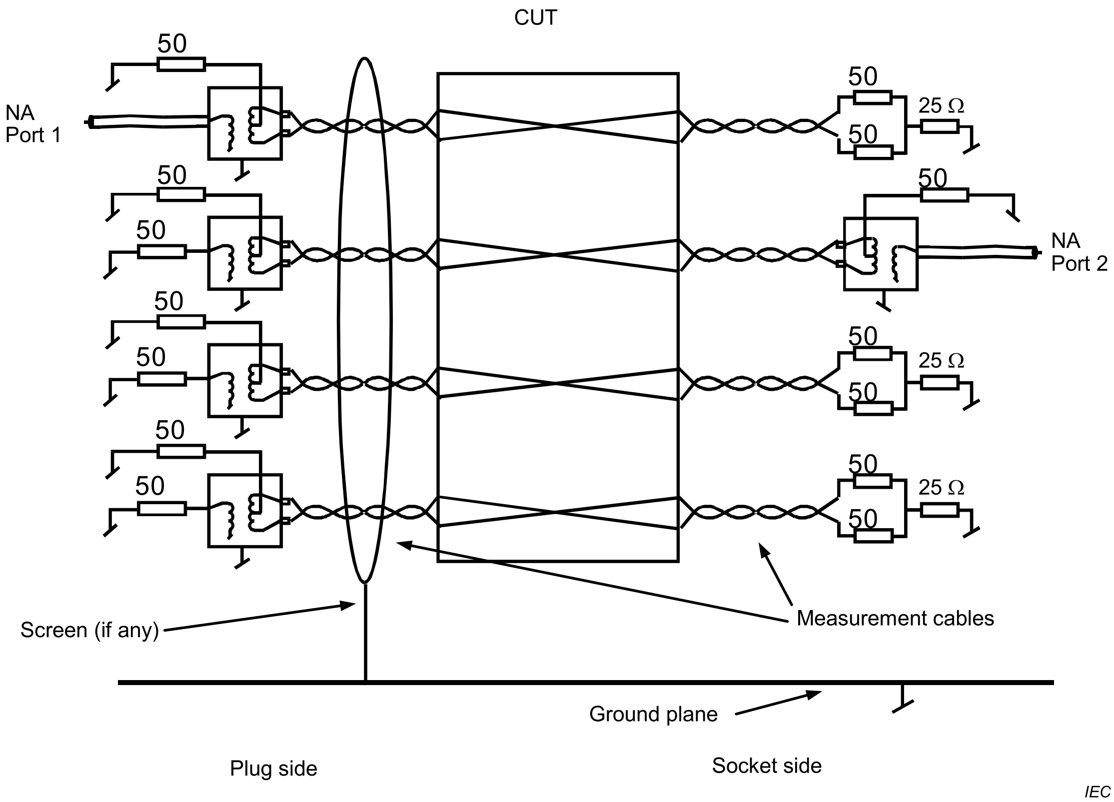
H.3 Test set-up

The test set-up consisting of two baluns and a network analyser shall be as given in Annex D. A diagram of the set-up, which also shows the termination principles, is shown in Figures H.1 and H.2.



Passive terminations shall be resistor terminations.

Figure H.1 – FEXT measurement differential mode only terminations



Passive terminations may be either balun or resistor terminations.

Figure H.2 – FEXT measurement differential and common mode terminations

H.4 Procedure

H.4.1 Calibration

Calibration shall be carried out as given in G.4.1.

H.4.2 Establishment of noise floor

The noise floor of the setup is established as shown in G.4.2.

H.5 Measurement

The disturbing pair of the CUT shall be connected to the signal source and the disturbed pair to the receiver port. Terminations shall be made in accordance with Figure H.1 and Figure H.2. All possible pair combinations[4] shall be tested and the results recorded.

It is recommended that the socket be terminated with short separated pairs without jacket. Test all possible pair combinations and record test results.

The CUT shall be tested in the following configurations:

a) With differential terminations only.

4 There are 12 different combinations for far end crosstalk in a four pair connector, which gives a total of 12 measurements for each termination method.

NOTE For category 7 this test is not required. This requirement is under consideration for all categories should experience show that a test for balance can replace this requirement.

b) With differential and common mode terminations.

H.6 Test report

The measured results shall be reported in graphical or table format with the specification limits shown on the graphs or in the table at the same frequencies as specified in the relevant detail specification. Results for all pairs shall be reported. It shall be explicitly noted if the measured results exceed the test limits.

H.7 Accuracy

The accuracy shall be better than ±1 dB at measurements up to 60 dB and ±2 dB at measurements up to 85 dB.

Annex I
(normative)

Transverse Conversion Loss (*TCL*) and Transverse Conversion Transfer Loss (*TCTL*)

I.1 Object

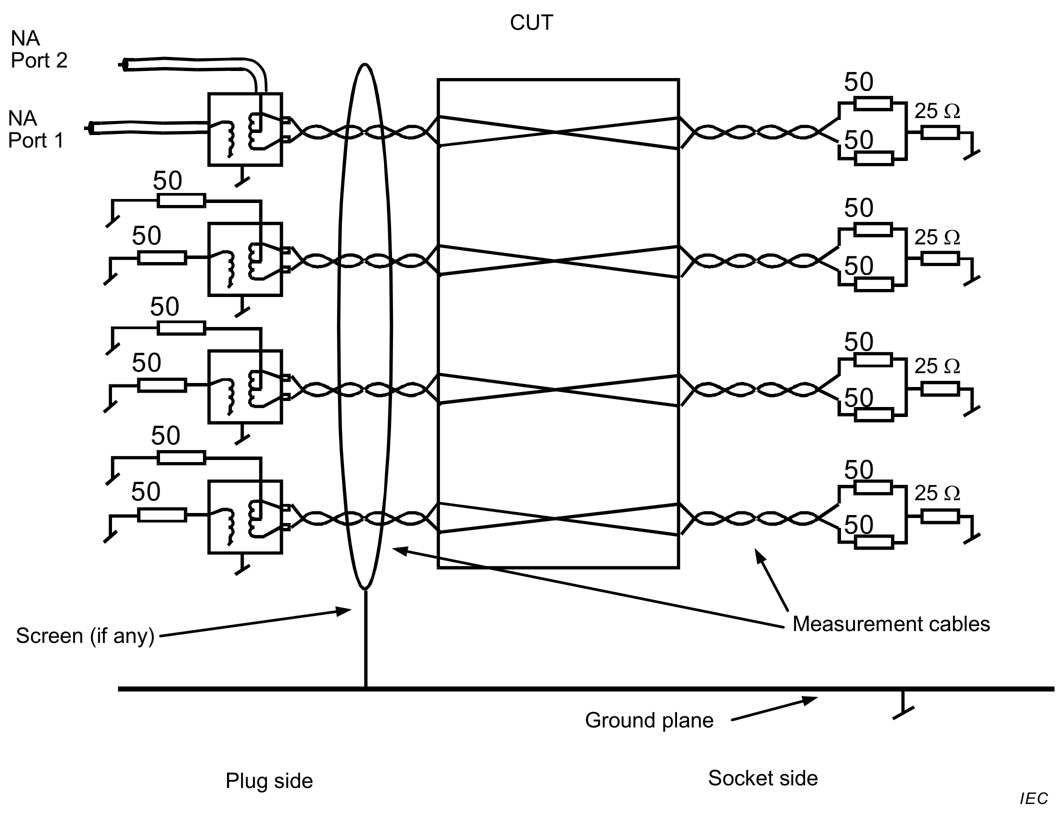
The object of this test shall be to measure the mode conversion (differential to common mode) of a signal in the conductor pairs of the CUT. This is also called unbalance attenuation or transverse conversion loss, *TCL*.

I.2 Test method

Balance shall be evaluated by measuring the common mode part of a differential mode signal, which is launched in one of the conductor pairs of the CUT.

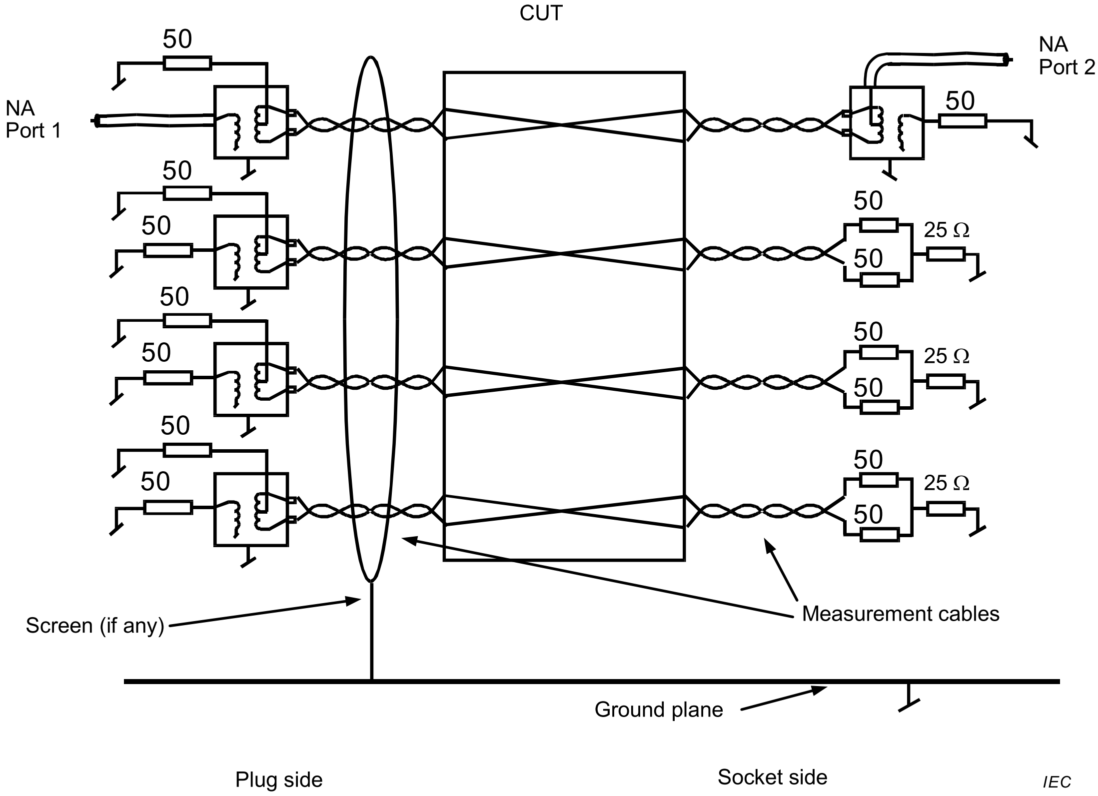
I.3 Test set-up

The test set-up shall comprise a network analyser and a balun with a differential and common mode test port. A representation of the set-up, which also shows the termination principles, is shown in Figure I.1 for *TCL* measurements and in Figure I.2 for *TCTL* measurements.



Passive terminations may be either balun or resistor terminations.

Figure I.1 – *TCL* measurement



Passive terminations may be either balun or resistor terminations.

Figure I.2 – *TCTL* measurement

I.4 Procedure

I.4.1 Calibration

Calibration shall be performed in three steps:

a) The attenuation of the coaxial test leads to the network analyser is calibrated out by performing a through calibration with these test leads connected together.

b) The attenuation of differential signals of the test balun, $a_{\mathrm{bal,DM}}$ is measured by connecting two identical baluns back to back. The insertion loss of these baluns is measured, and half of this loss is the insertion loss of the balun for a differential signal.

c) The attenuation of common mode signals of the test balun, $a_{\mathrm{bal,CM}}$ is measured by measuring the insertion loss from the common mode test port of the balun to the differential output terminals. The two differential output terminals shall be short-circuited and connected to the inner conductor of the coaxial test lead to the network analyser.

I.4.2 Noise floor

The noise floor of the set-up shall be measured. The level of the noise floor is determined by white noise, which may be reduced by increasing the test power and by reducing the bandwidth of the network analyser, and by the longitudinal balance (see Table D.1) of the test balun.

The noise floor, $a_{\mathrm{noise,m}}$ shall be measured by terminating the differential output of the balun with a 100 Ω resistor and perform a S_{21} measurement between the differential mode and common mode test port of the balun. a_{noise} is calculated as:

$$a_{\mathrm{noise,m}} = -20\log\left|S_{21}\right|$$

$$a_{\text{noise}} = a_{\text{noise,m}} - a_{\text{bal,DM}} - a_{\text{bal,CM}}$$

The noise floor shall be 20 dB lower than any specified limit for balance. If the measured value is closer to the noise floor than 10 dB, this shall be reported.

I.4.3 Measurement

The S_{21} measurement shall be carried out as follows: connect the measured pair of the CUT to the differential output of the test balun. Terminate the CUT according to Clause D.6. The balance, TCL is calculated as:

$$a_{\text{meas}} = -20\log\left|S_{21}\right|$$

$$TCL \text{ or } TCTL = a_{\text{meas}} - a_{\text{bal,DM}} - a_{\text{bal,CM}}$$

I.5 Test report

The measured results shall be reported in graphical or table format with the specification limits shown on the graphs or in the table at the same frequencies as specified in the relevant detail specification. Results for all pairs shall be reported. It shall be explicitly noted if the measured results exceed the test limits.

I.6 Accuracy

The accuracy shall be better than ±1 dB at the specification limit.

Annex J
(normative)

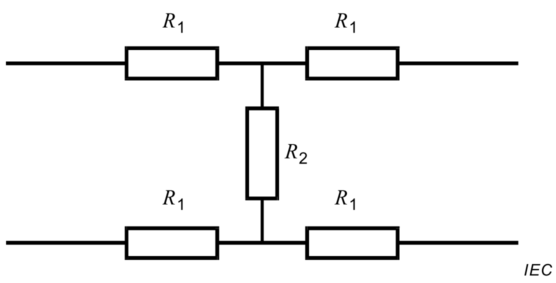
Termination of balun

J.1 Termination of balun with low return loss for common mode

If the available balun does not provide common mode termination (centre tap is either connected to ground or open), a balanced resistor attenuator shall be applied in order to provide the required return loss. The attenuator shall be implemented at a small printed circuit board mounted with SMD resistors. There are two cases: one for the centre tap connected to ground and one for the centre tap open.

J.2 Centre tap connected to ground

The attenuator used shall be as shown at Figure J.1. The nominal attenuation is 10 dB and the calculated common mode impedance is 26 Ω.



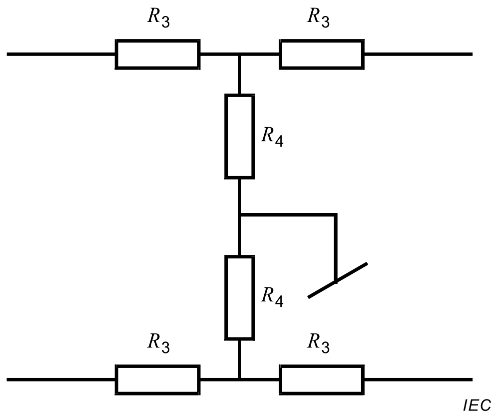
R_1 = 26 Ω

R_2 = 70 Ω

Figure J.1 – Balanced attenuator for balun centre tap grounded

J.3 Centre tap open

The attenuator used shall be as shown at Figure J.2. The nominal attenuation is 5 dB and the calculated common mode impedance is 48 Ω.



R_3 = 14 Ω

R_4 = 82 Ω

NOTE Resistor values are nominal. The nearest standard values can be chosen.

Figure J.2 – Balanced attenuator for balun centre tap open

Bibliography

IEC 60068-2-38, *Environmental testing – Part 2-38: Tests – Test Z/AD: Composite temperature/humidity cyclic test*

IEC 60512-25-9, *Connectors for electronic equipment – Tests and measurements – Part 25-9: Signal integrity tests –Test 25i: Alien crosstalk*

IEC 60603-7, *Connectors for electronic equipment – Part 7: Detail specification for 8-way, unshielded, free and fixed connectors*

IEC 61196 (all parts), *Coaxial communication cables*

IEC 62153-4-12, *Metallic communication cable test methods – Part 4-12: Electromagnetic compatibility (EMC) –Coupling attenuation or screening attenuation of connecting hardware – Absorbing clamp method*

Recommendation ITU-T G.117, *Transmission systems and media – General characteristics of international telephone connections and international telephone circuits – Transmission aspects of unbalance about earth*

Recommendation ITU-T K.20, *Resistibility of telecommunication equipment installed in a telecommunication centre to overvoltages and overcurrents*

Recommendation ITU-T O.9, *Series O: Specifications of measuring equipment – Measuring arrangements to assess the degree of unbalance about earth*

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